수위 예측 AI
성능 개선 결과 4

1. 누적 강우 110mm 이상

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

학습데이터

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우 110mm 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 11개 中 9개 데이터 파일 학습에 사용 (검증 2개)

검증 : 학습 데이터에서 제외한 2개 + 4개 이벤트 사용

- 학습 : 2014 ~ 2022 (7개)
- 검증 : 2023(1개)
- 테스트 : 2024 (1개)

* model_(gn/dg)_110

누적 강우 110mm 이상 예측 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

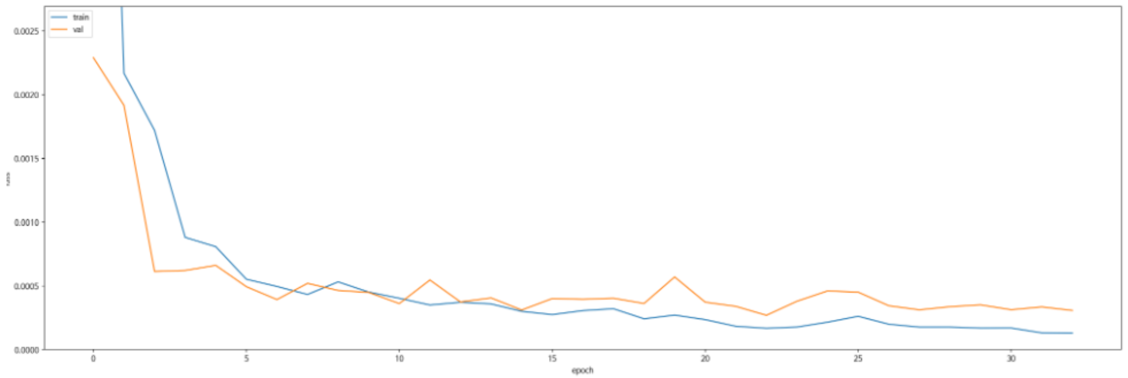
일괄 예측 수행 결과

Train (1785 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0370	0.0225	0.0501	0.992969	0.993190	2.751991

Valid (228 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0489	0.0280	0.0646	0.958571	0.960290	2.528366

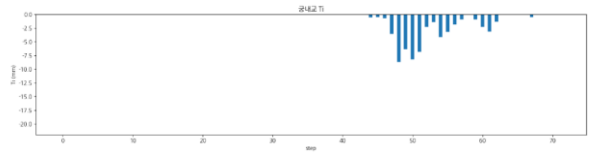
Test (25 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0698	0.0330	0.0905	0.778608	0.844471	10.796319

Train Chart

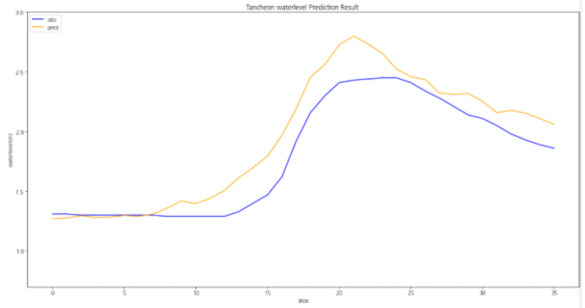


누적 강우 110mm 이상
예측 코드 – 궁내교 1

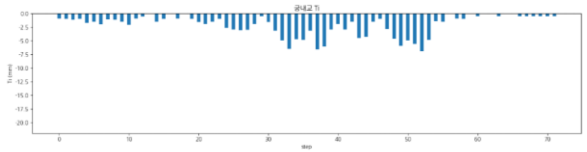
	궁내교
testdata_200802	r2 : 0.8203 mse : 0.0379 mae : 0.1586 rmse : 0.1949
testdata_220712	r2 : -0.3240 mse : 0.0525 mae : 0.1927 rmse : 0.2292
testdata_230711	r2 : 0.6352 mse : 0.1020 mae : 0.2347 rmse : 0.3194



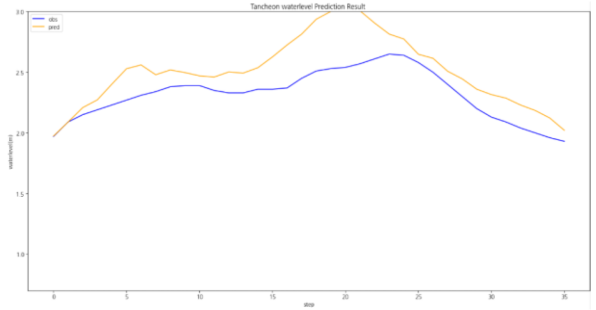
testdata_ 200802



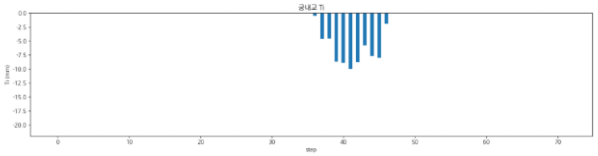
testdata_ 200802



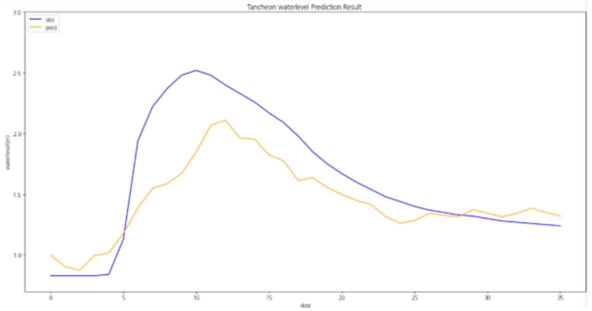
testdata_ 220712



testdata_ 220712



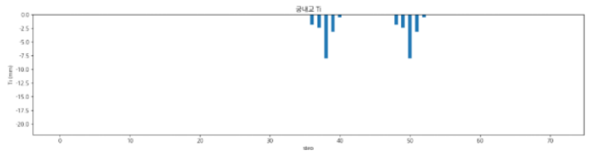
testdata_230711



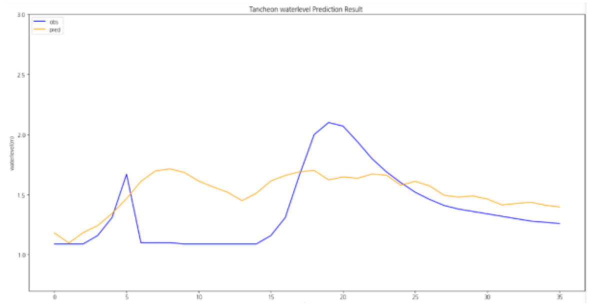
testdata_230711

누적 강우 110mm 이상
예측 코드 – 궁내교 2

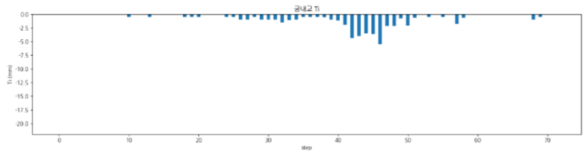
	궁내교
testdata_240723	r2 : -0.0433 mse : 0.0972 mae :0.2456 rmse : 0.3118
testdata_250716	r2 : 0.6714 mse : 0.0329 mae : 0.1470 rmse : 0.1815
testdata_250814	r2 : 0.6227 mse : 0.0265 mae : 0.1210 rmse : 0.1630



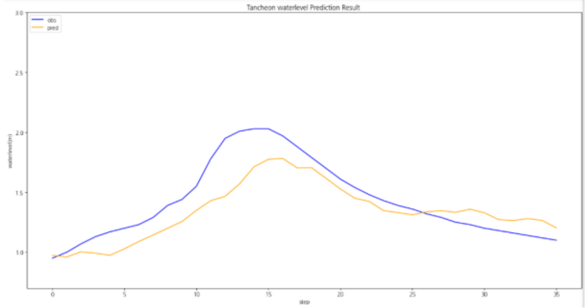
testdata_ 240723



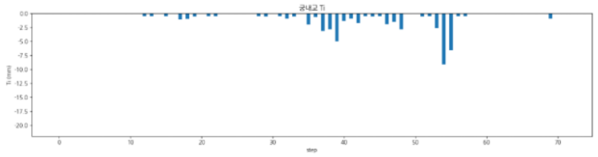
testdata_ 240723



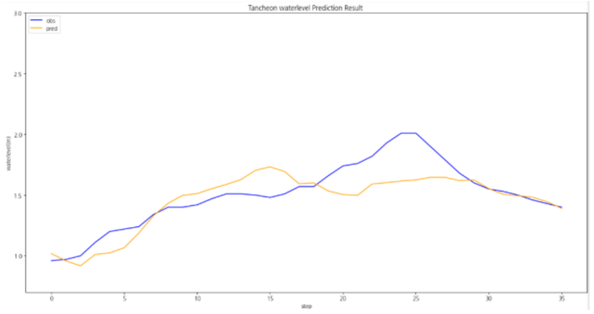
testdata_ 250716



testdata_ 250716



testdata_ 250814



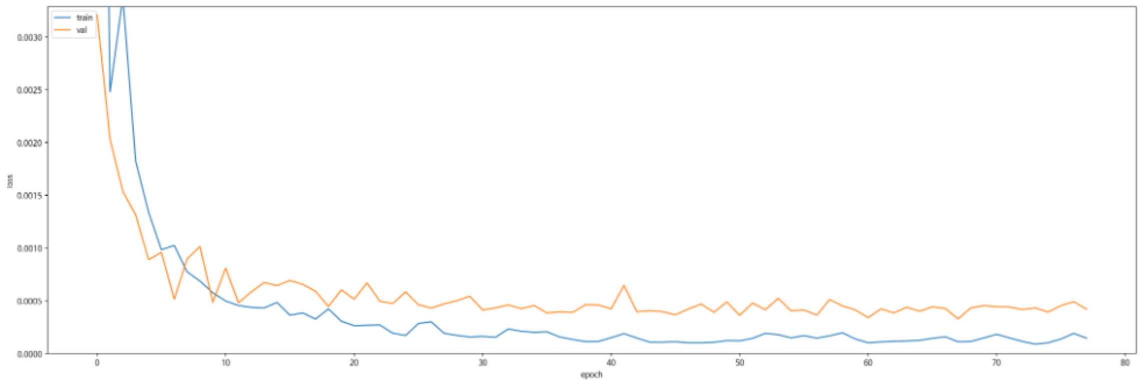
testdata_250814

* model_(gn/dg)_110

누적 강우 110mm 이상 예측 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

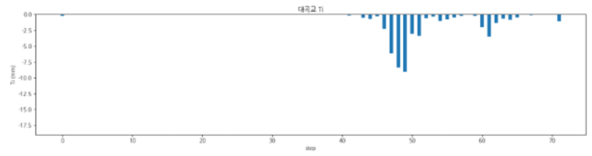
Train (1785 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0451	0.0139	0.0689	0.996132	0.996789	3.457385

Valid (228 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0818	0.0253	0.1113	0.963230	0.965334	3.448578

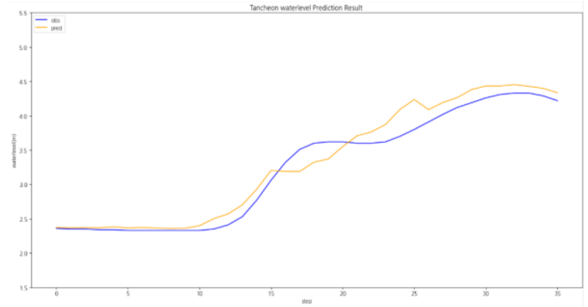
Test (25 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.1498	0.0398	0.1888	0.663938	0.766153	4.633667

누적 강우 110mm 이상
예측 코드 – 대곡교 1

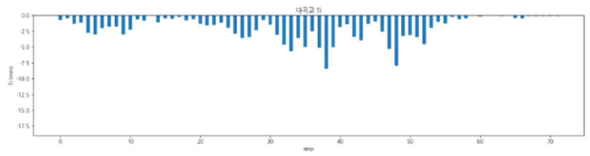
	대곡교
testdata_200802	r2 : 0.9516 mse : 0.0295 mae : 0.1382 rmse : 0.1717
testdata_220712	r2 : -0.6522 mse : 0.2313 mae : 0.4550 rmse : 0.4810
testdata_230711	r2 : 0.8627 mse : 0.1249 mae : 0.2790 rmse : 0.3534



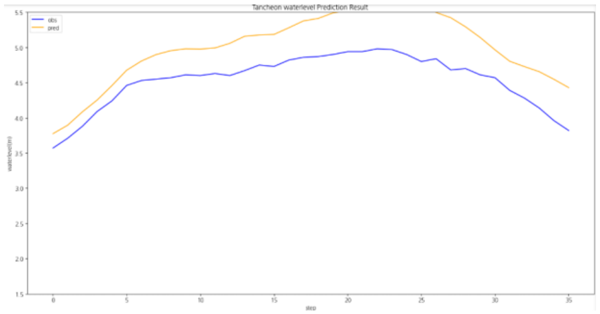
testdata_200802



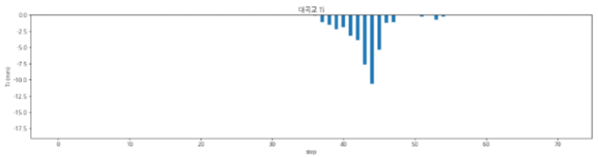
testdata_ 200802



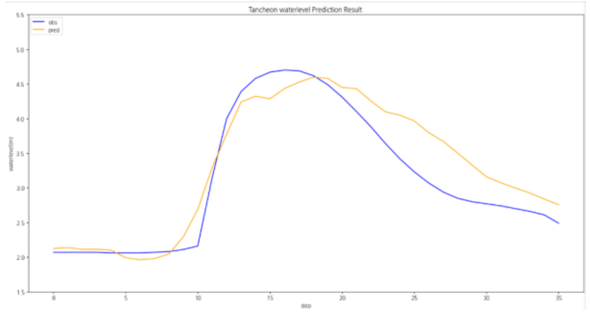
testdata_ 220712



testdata_ 220712



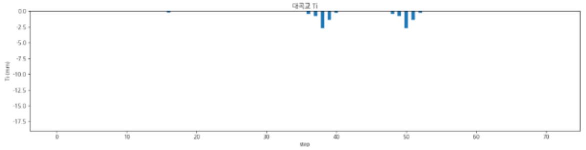
testdata_ 230711



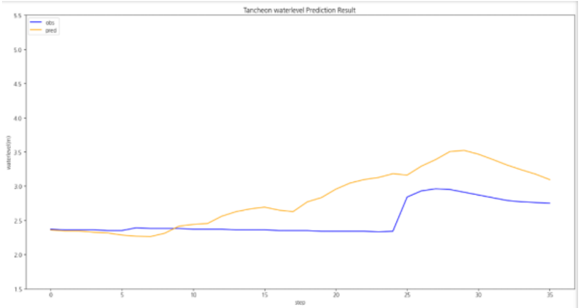
testdata_ 230711

누적 강우 110mm 이상
예측 코드 – 대곡교 2

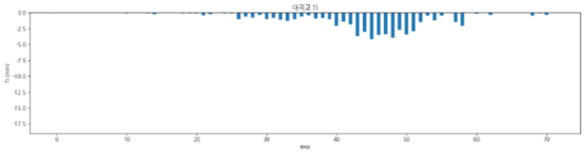
	대곡교
testdata_240723	r2 : -2.2679 mse : 0.1745 mae : 0.3375 rmse : 0.4177
testdata_250716	r2 : 0.9700 mse : 0.0138 mae : 0.0916 rmse : 0.1176
testdata_250814	r2 : 0.7924 mse : 0.0720 mae : 0.2086 rmse : 0.2684



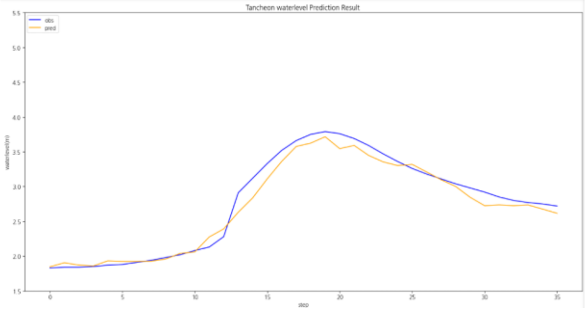
testdata_240723



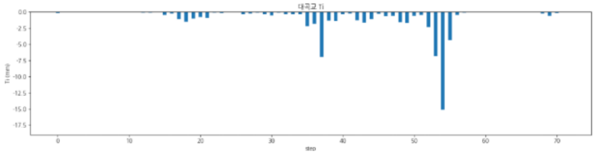
testdata_240723



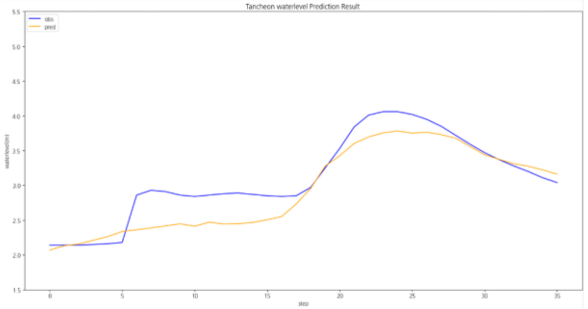
testdata_250716



testdata_250716



testdata_250814



testdata_250814

* model_(gn/dg)_110_2

1-2.누적 강우 110mm 이상 조합 찾기

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우 110mm 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

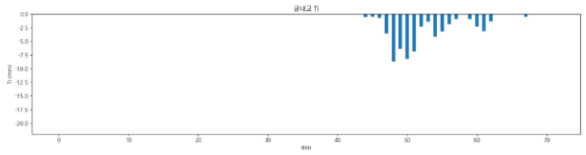
학습 : 총 11개 中 9개 데이터 파일 학습에 사용 (검증 2개)

검증 : 학습 데이터에서 제외한 2개 + 4개 이벤트 사용

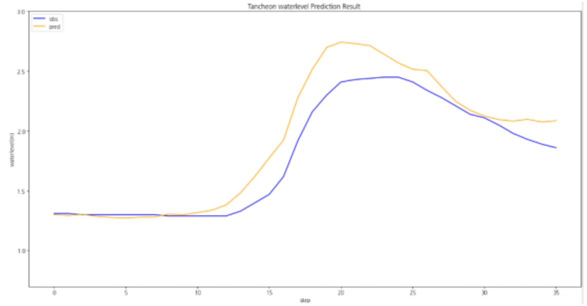
- 학습 : 2017 ~ 2023 (8개)
- 검증 : 2024(1개)
- 테스트 : 2014 (1개)

누적 강우 110mm 이상 조합 찾기 2
예측 코드 – 궁내교 1

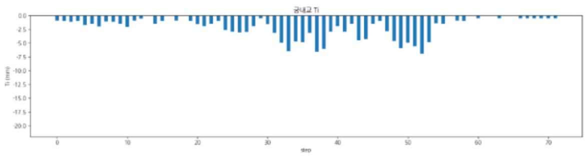
	궁내교
testdata_200802	r2 : 0.8432 mse : 0.0331 mae : 0.1339 rmse : 0.1821
testdata_220712	r2 : 0.0759 mse : 0.0366 mae : 0.1551 rmse : 0.1914
testdata_230711	r2 : 0.8378 mse : 0.0453 mae : 0.1678 rmse : 0.2130



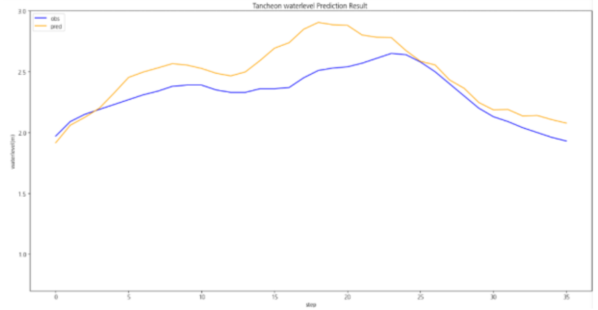
testdata_ 200802



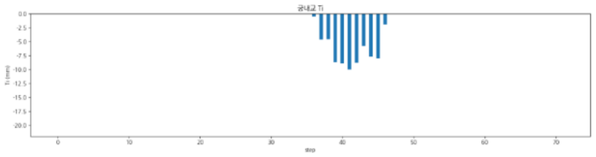
testdata_ 200802



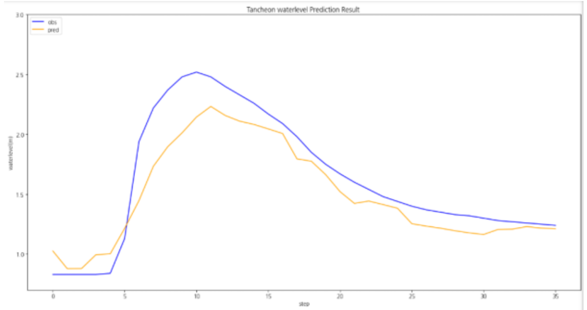
testdata_ 220712



testdata_ 220712



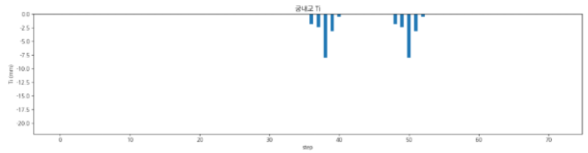
testdata_230711



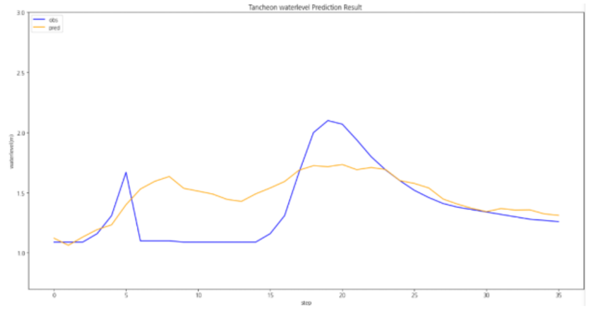
testdata_230711

누적 강우 110mm 이상 조합 찾기 2
예측 코드 – 궁내교 2

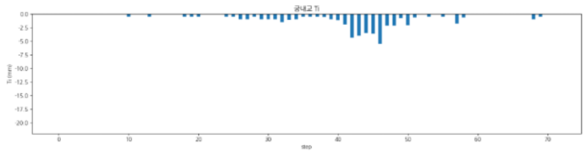
	궁내교
testdata_240723	r2 : 0.2846 mse : 0.0666 mae : 0.1896 rmse : 0.2581
testdata_250716	r2 : 0.3038 mse : 0.0698 mae : 0.1956 rmse : 0.2642
testdata_250814	r2 : 0.7643 mse : 0.0166 mae : 0.0963 rmse : 0.1289



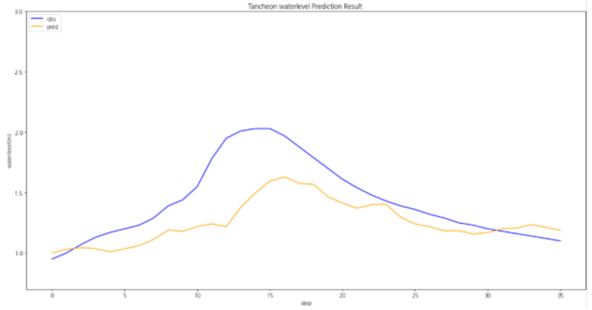
testdata_ 240723



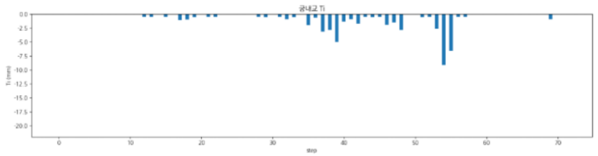
testdata_ 240723



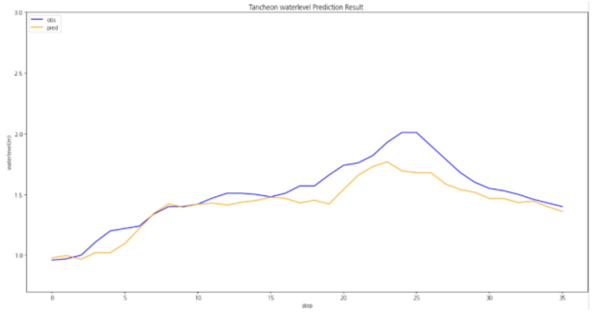
testdata_ 250716



testdata_ 250716



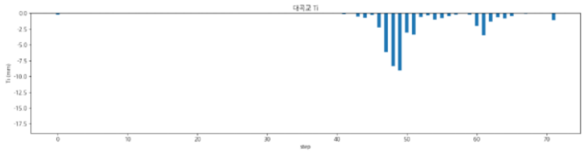
testdata_ 250814



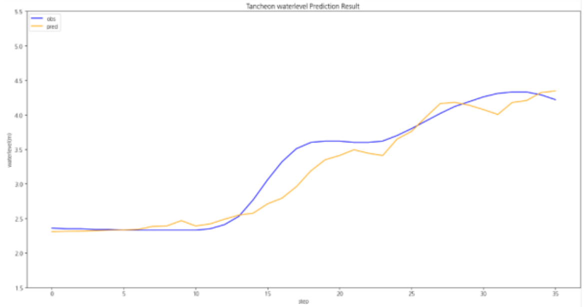
testdata_250814

누적 강우 110mm 이상 조합 찾기 2
예측 코드 – 대곡교 1

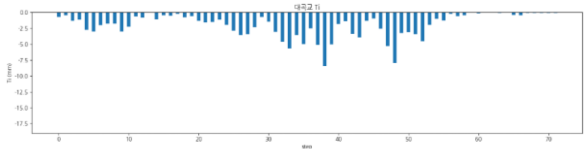
	대곡교
testdata_200802	r2 : 0.9377 mse : 0.0380 mae : 0.1374 rmse : 0.1949
testdata_220712	r2 : 0.1093 mse : 0.1247 mae : 0.3037 rmse : 0.3531
testdata_230711	r2 : 0.9085 mse : 0.0832 mae : 0.2344 rmse : 0.2885



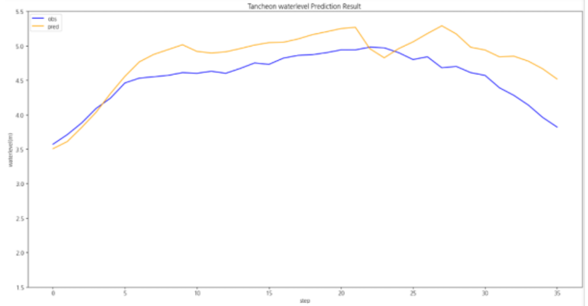
testdata_200802



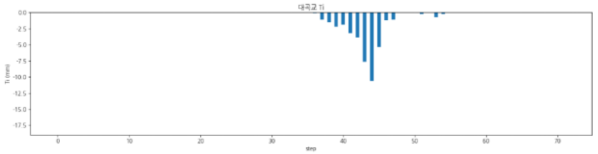
testdata_ 200802



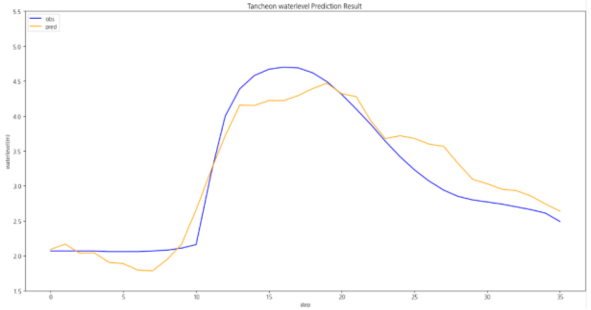
testdata_ 220712



testdata_ 220712



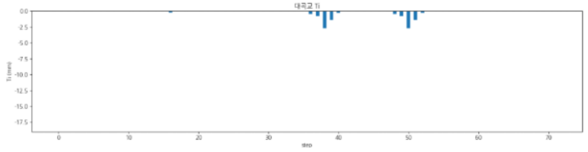
testdata_ 230711



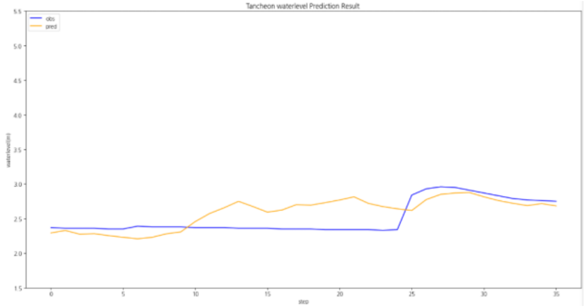
testdata_ 230711

누적 강우 110mm 이상 조합 찾기 2
예측 코드 – 대곡교 2

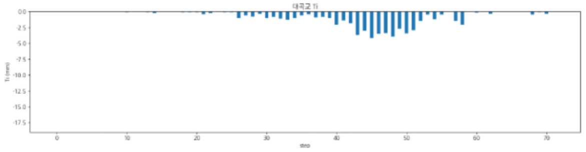
	대곡교
testdata_240723	r2 : 0.0119 mse : 0.0527 mae : 0.1883 rmse : 0.2297
testdata_250716	r2 : 0.8820 mse : 0.0544 mae : 0.1842 rmse : 0.2334
testdata_250814	r2 : 0.6907 mse : 0.1073 mae : 0.2570 rmse : 0.3276



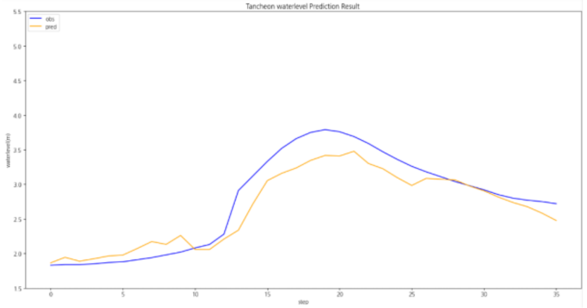
testdata_240723



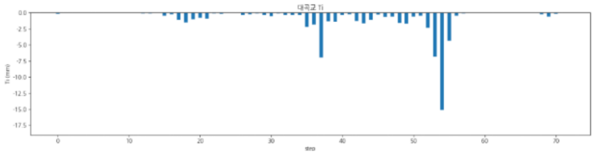
testdata_240723



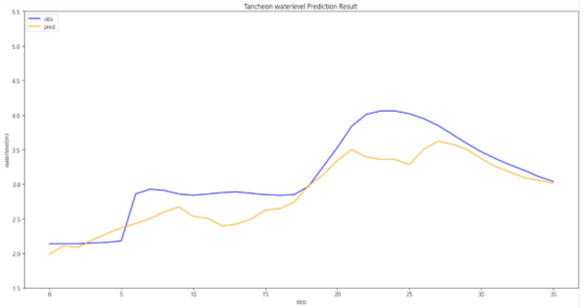
testdata_250716



testdata_250716



testdata_250814



testdata_250814

* model_(gn/dg)_110_3

1-3. 누적 강우 110mm 이상 조합 찾기

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우 110mm 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

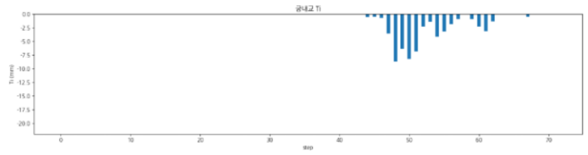
학습 : 총 11개 中 9개 데이터 파일 학습에 사용 (검증 2개)

검증 : 학습 데이터에서 제외한 2개 + 4개 이벤트 사용

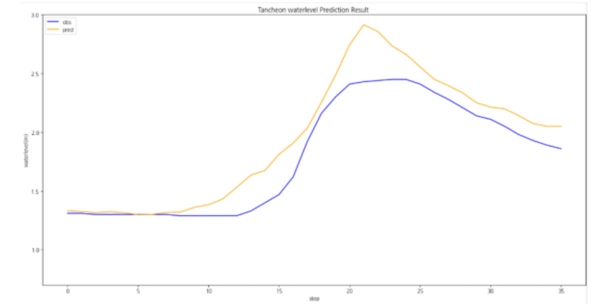
- 학습 : 2014 ~ 2023 (8개)
- 검증 : 2022(6667) (1개)
- 테스트 : 2024 (1개)

누적 강우 110mm 이상 조합 찾기 3
예측 코드 – 궁내교 1

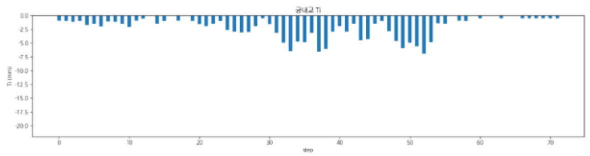
	궁내교
testdata_200802	r2 : 0.8183 mse : 0.0384 mae : 0.1541 rmse : 0.1960
testdata_220712	r2 : -0.3856 mse : 0.0549 mae : 0.2168 rmse : 0.2344
testdata_230711	r2 : 0.6955 mse : 0.0851 mae : 0.2246 rmse : 0.2918



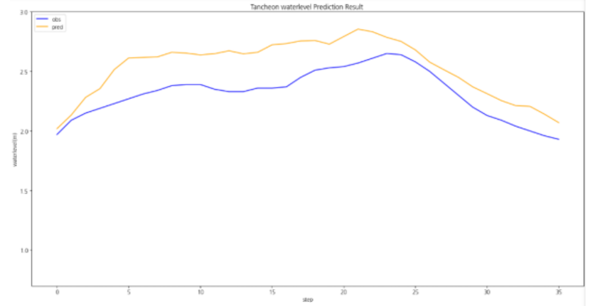
testdata_ 200802



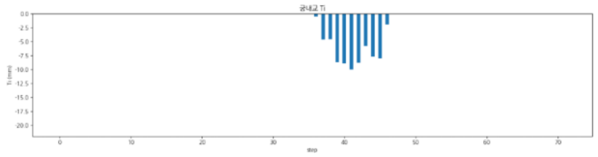
testdata_ 200802



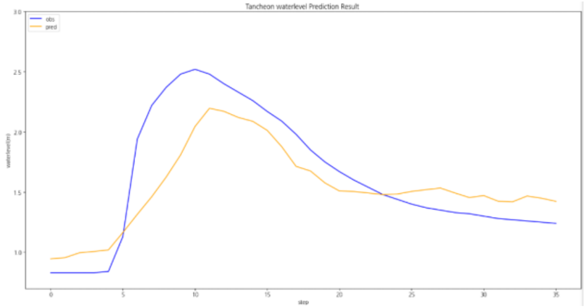
testdata_ 220712



testdata_ 220712



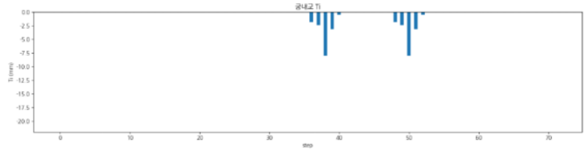
testdata_230711



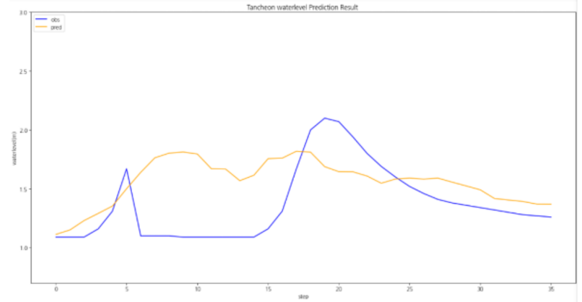
testdata_230711

누적 강우 110mm 이상 조합 찾기 3
예측 코드 – 궁내교 2

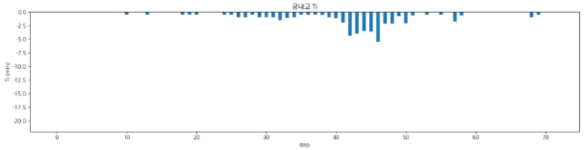
	궁내교
testdata_240723	r2 : -0.4278 mse : 0.1330 mae : 0.2866 rmse : 0.3647
testdata_250716	r2 : 0.7261 mse : 0.0274 mae : 0.1434 rmse : 0.1657
testdata_250814	r2 : 0.8484 mse : 0.0106 mae : 0.0913 rmse : 0.1033



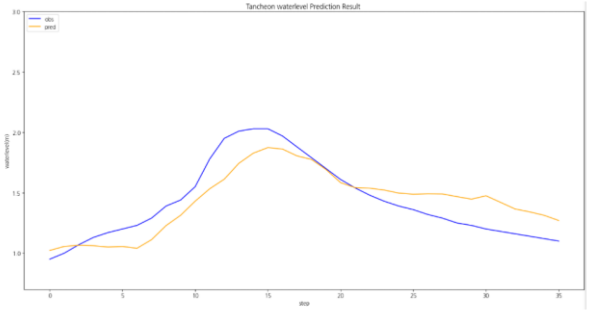
testdata_ 240723



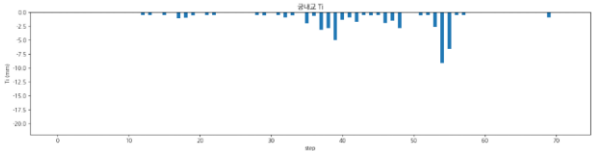
testdata_ 240723



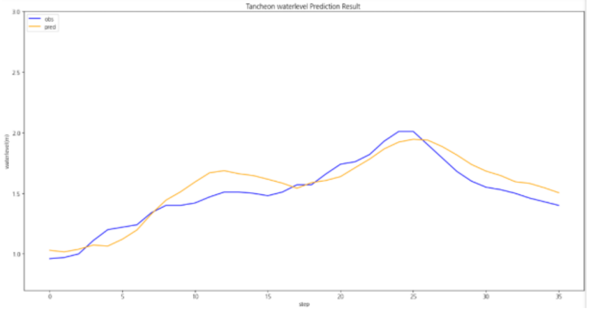
testdata_ 250716



testdata_ 250716



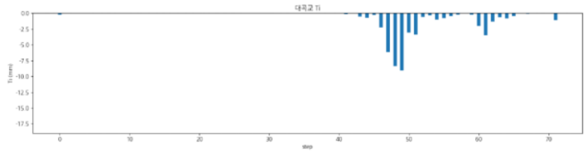
testdata_ 250814



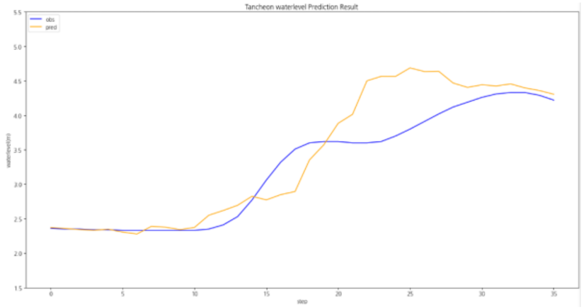
testdata_250814

누적 강우 110mm 이상 조합 찾기 3
예측 코드 – 대곡교 1

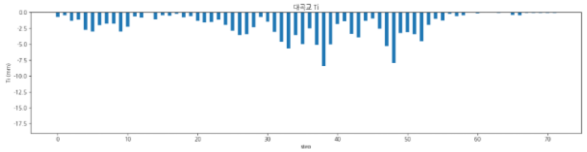
	대곡교
testdata_200802	r2 : 0.7493 mse : 0.1529 mae : 0.2611 rmse : 0.3910
testdata_220712	r2 : -1.6344 mse : 0.3689 mae : 0.5123 rmse : 0.6073
testdata_230711	r2 : 0.7544 mse : 0.2235 mae : 0.3753 rmse : 0.4727



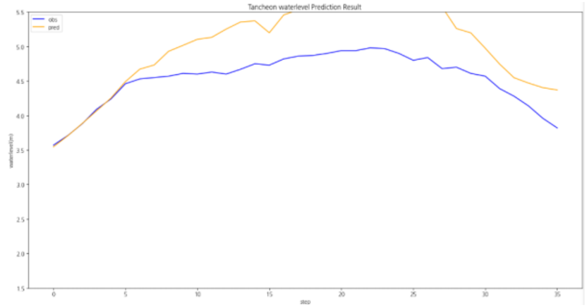
testdata_200802



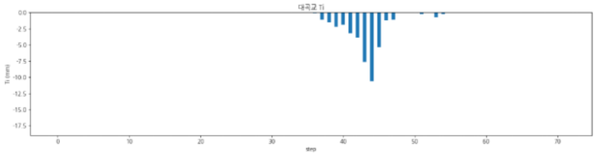
testdata_ 200802



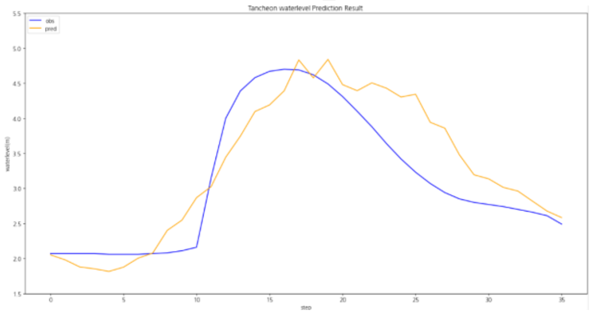
testdata_ 220712



testdata_ 220712



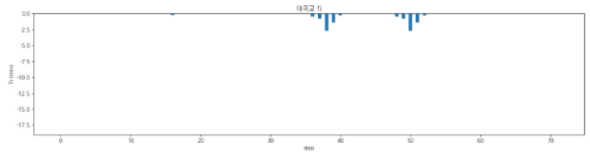
testdata_ 230711



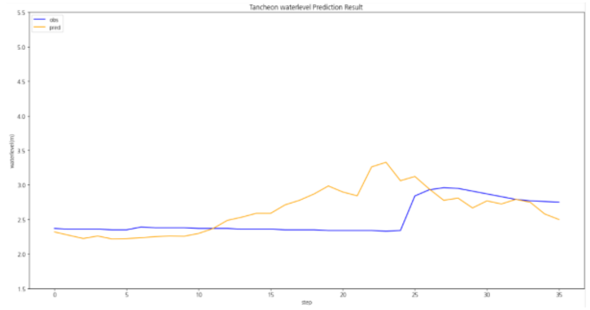
testdata_ 230711

누적 강우 110mm 이상 조합 찾기 3
예측 코드 – 대곡교 2

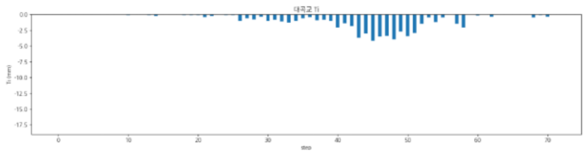
	대곡교
testdata_240723	r2 : -1.3562 mse : 0.1258 mae : 0.2542 rmse : 0.3547
testdata_250716	r2 : 0.8936 mse : 0.0490 mae : 0.1713 rmse : 0.2215
testdata_250814	r2 : 0.7912 mse : 0.0724 mae : 0.2211 rmse : 0.2691



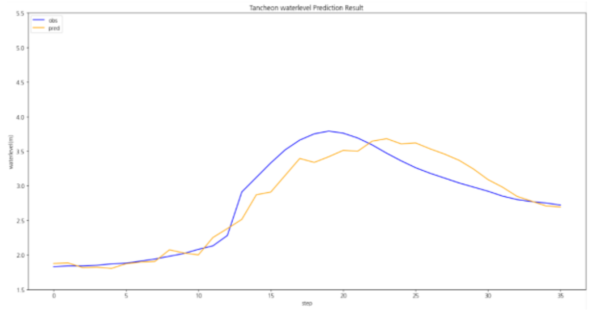
testdata_240723



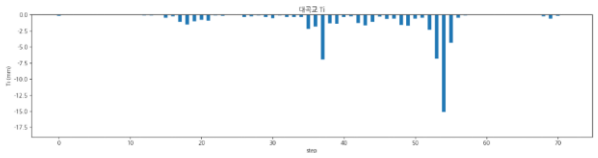
testdata_240723



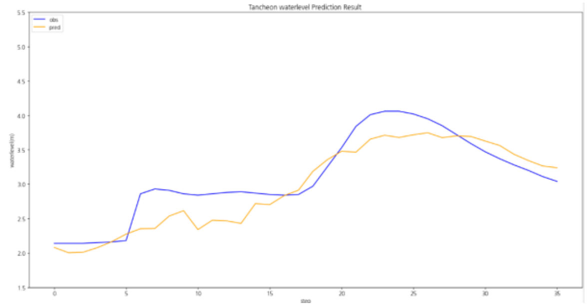
testdata_250716



testdata_250716



testdata_250814



testdata_250814

* model_(gn/dg)_110_4

1-4. 누적 강우 110mm 이상 조합 찾기

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우 110mm 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

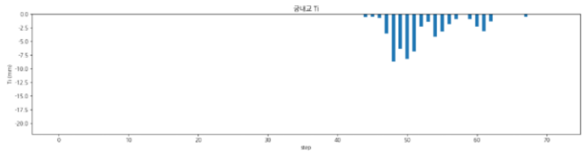
학습 : 총 11개 中 9개 데이터 파일 학습에 사용 (검증 2개)

검증 : 학습 데이터에서 제외한 2개 + 4개 이벤트 사용

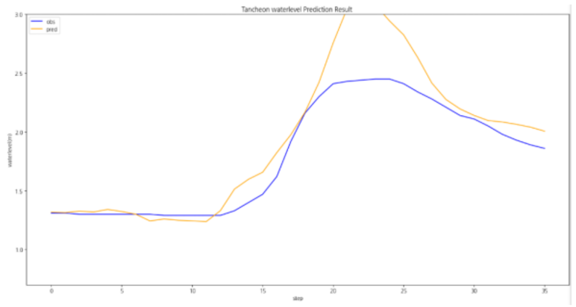
- 학습 : 2014 ~ 2024 (8개)
- 검증 : 2022(39) (1개)
- 테스트 : 2022(6667) (1개)

누적 강우 110mm 이상 조합 찾기 4
예측 코드 – 궁내교 1

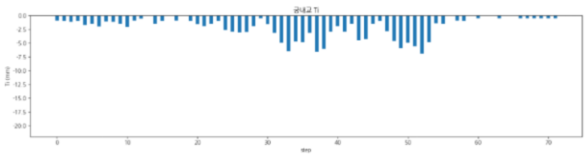
	궁내교
testdata_200802	r2 : 0.7237 mse : 0.0584 mae : 0.1572 rmse : 0.2417
testdata_220712	r2 : 0.3192 mse : 0.0270 mae : 0.1369 rmse : 0.1643
testdata_230711	r2 : 0.8533 mse : 0.0410 mae : 0.1509 rmse : 0.2025



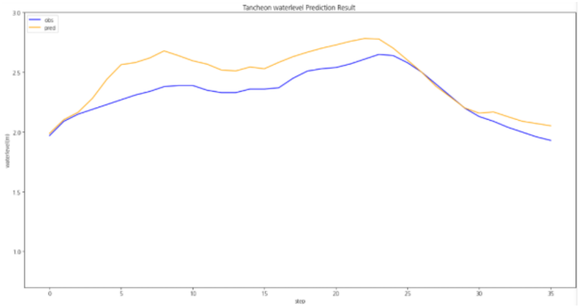
testdata_ 200802



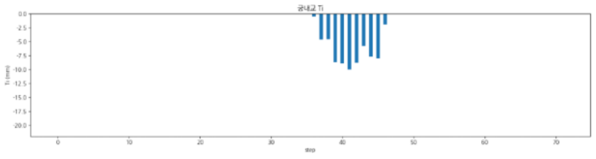
testdata_ 200802



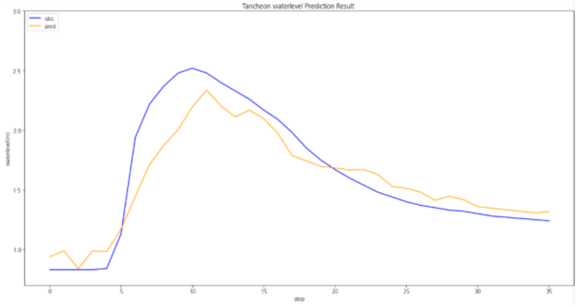
testdata_ 220712



testdata_ 220712



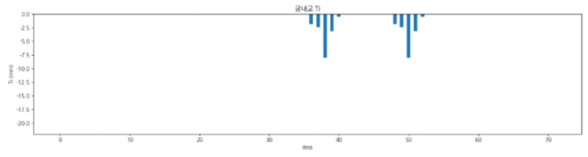
testdata_230711



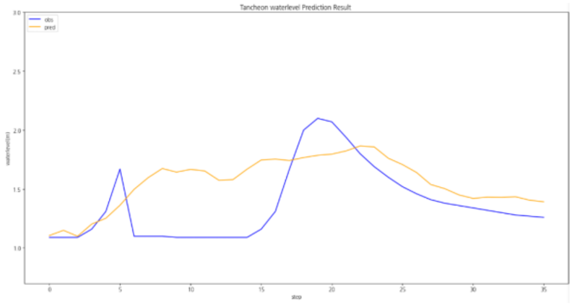
testdata_230711

누적 강우 110mm 이상 조합 찾기 4
예측 코드 – 궁내교 2

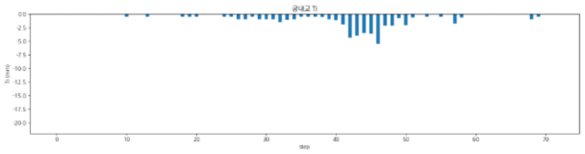
	궁내교
testdata_240723	r2 : -0.0894 mse : 0.1015 mae : 0.2527 rmse : 0.3186
testdata_250716	r2 : 0.8108 mse : 0.0189 mae : 0.1236 rmse : 0.1377
testdata_250814	r2 : 0.8475 mse : 0.0107 mae : 0.0815 rmse : 0.1036



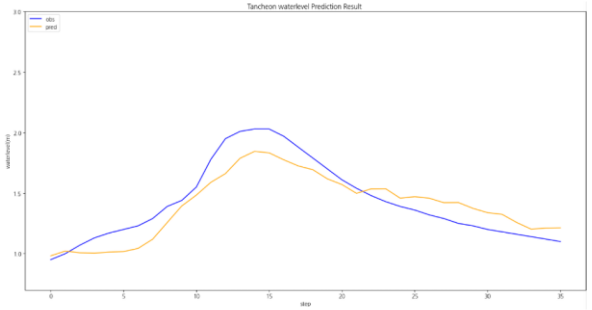
testdata_ 240723



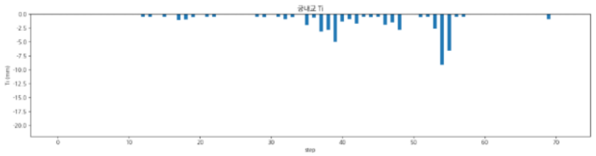
testdata_ 240723



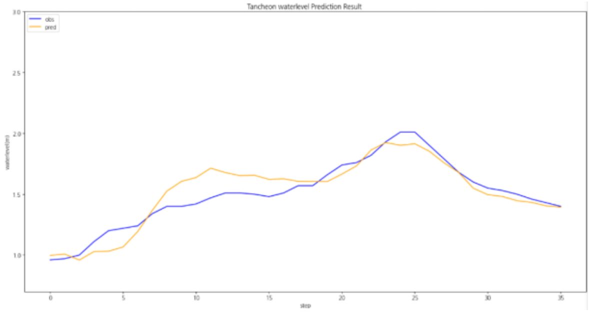
testdata_ 250716



testdata_ 250716



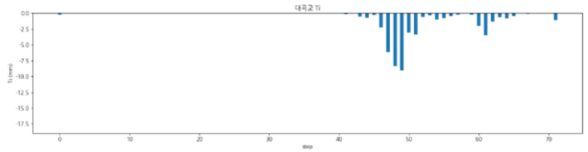
testdata_ 250814



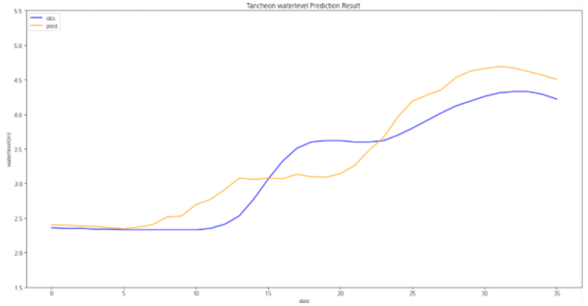
testdata_250814

누적 강우 110mm 이상 조합 찾기 4
예측 코드 – 대곡교 1

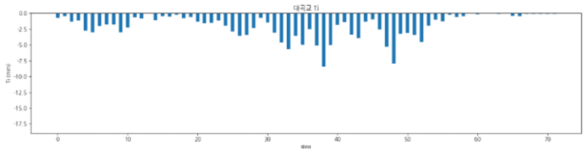
	대곡교
testdata_200802	r2 : 0.8350 mse : 0.1006 mae : 0.2683 rmse : 0.3172
testdata_220712	r2 : -0.0424 mse : 0.1459 mae : 0.3414 rmse : 0.3820
testdata_230711	r2 : 0.7456 mse : 0.2314 mae : 0.3847 rmse : 0.4811



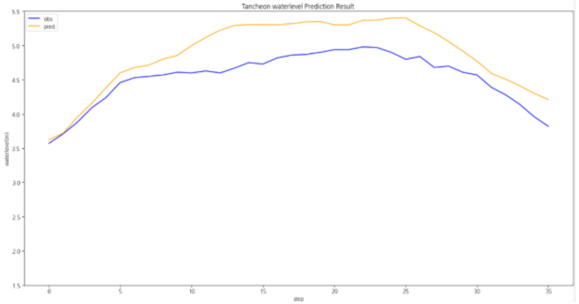
testdata_200802



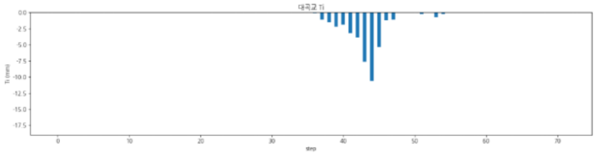
testdata_ 200802



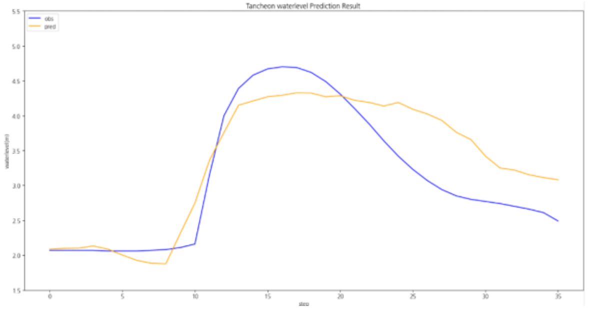
testdata_ 220712



testdata_ 220712



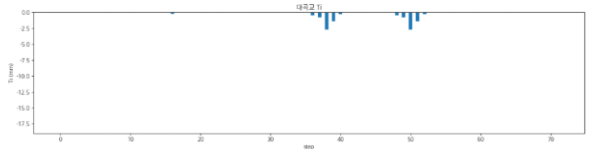
testdata_ 230711



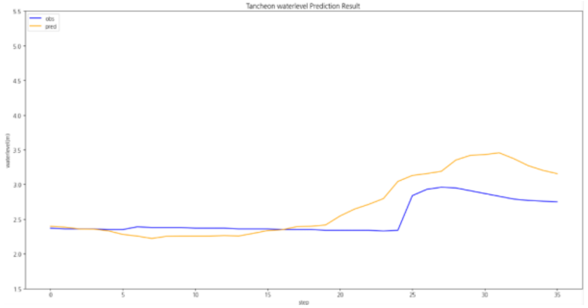
testdata_ 230711

누적 강우 110mm 이상 조합 찾기 4
예측 코드 – 대곡교 2

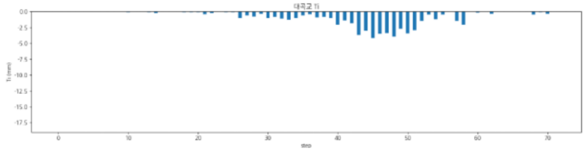
	대곡교
testdata_240723	r2 : -0.7705 mse : 0.0945 mae : 0.2284 rmse : 0.3075
testdata_250716	r2 : 0.8305 mse : 0.0782 mae : 0.1994 rmse : 0.2797
testdata_250814	r2 : 0.5011 mse : 0.1731 mae : 0.3295 rmse : 0.4160



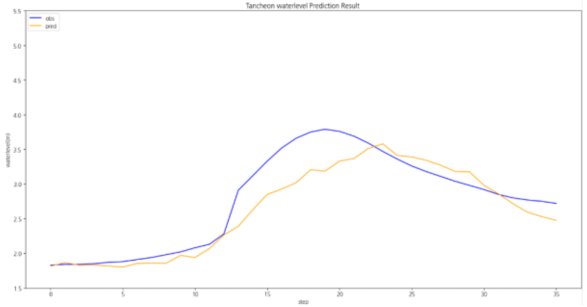
testdata_240723



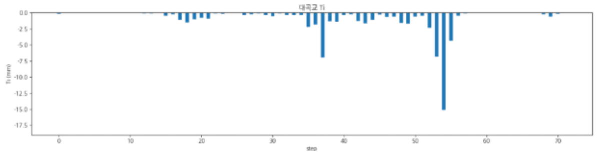
testdata_240723



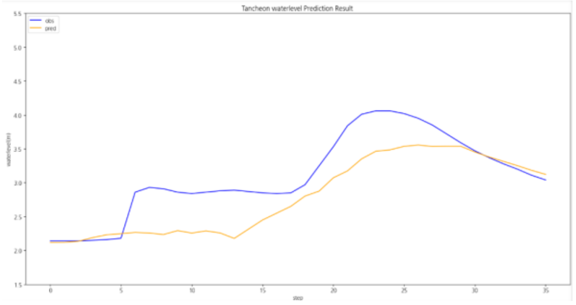
testdata_250716



testdata_250716



testdata_250814



testdata_250814

* model_(gn/dg)_110_5

1-5. 누적 강우 110mm 이상 조합 찾기

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우 110mm 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

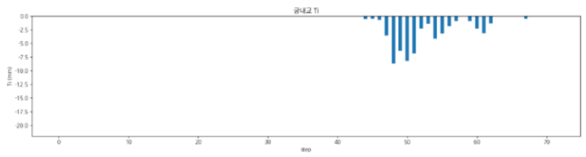
학습 : 총 11개 中 9개 데이터 파일 학습에 사용 (검증 2개)

검증 : 학습 데이터에서 제외한 2개 + 4개 이벤트 사용

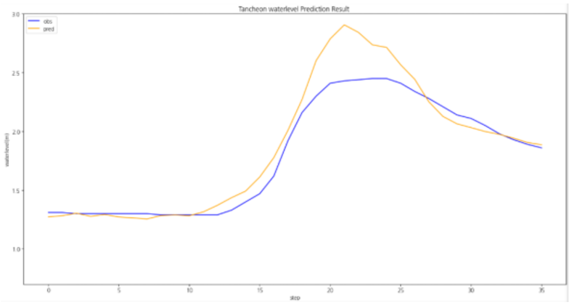
- 학습 : 2014 ~ 2024 (8개)
- 검증 : 2022(35) (1개)
- 테스트 : 2022(6667) (1개)

누적 강우 110mm 이상 조합 찾기 5
예측 코드 – 궁내교 1

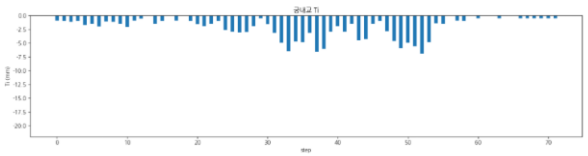
	궁내교
testdata_200802	r2 : 0.8774 mse : 0.0259 mae : 0.1047 rmse : 0.1609
testdata_220712	r2 : 0.0586 mse : 0.0373 mae : 0.1603 rmse : 0.1932
testdata_230711	r2 : 0.8028 mse : 0.0851 mae : 0.1455 rmse : 0.2349



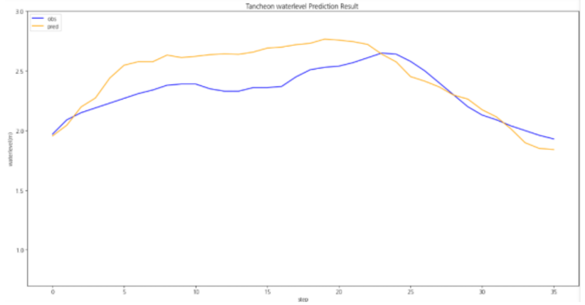
testdata_ 200802



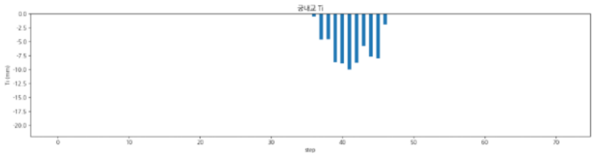
testdata_ 200802



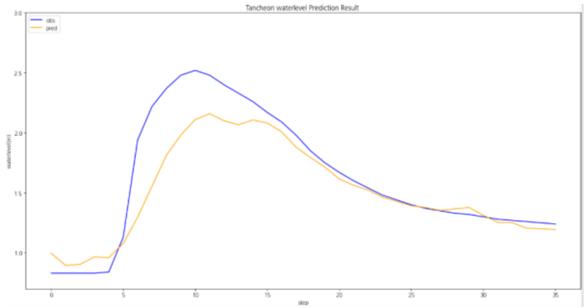
testdata_ 220712



testdata_ 220712



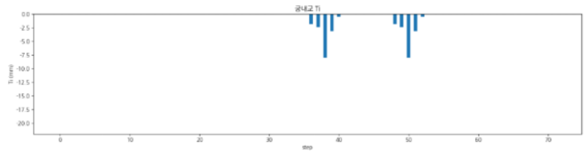
testdata_230711



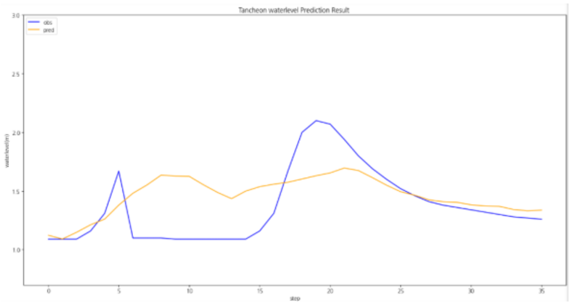
testdata_230711

누적 강우 110mm 이상 조합 찾기 5
예측 코드 – 궁내교 2

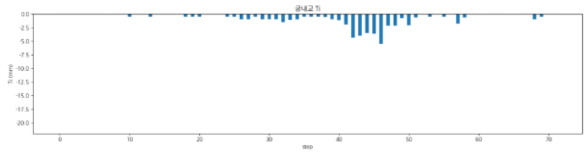
	궁내교
testdata_240723	r2 : 0.1532 mse : 0.0789 mae : 0.2090 rmse : 0.2809
testdata_250716	r2 : 0.7764 mse : 0.0224 mae : 0.1110 rmse : 0.1497
testdata_250814	r2 : 0.8673 mse : 0.0093 mae : 0.0756 rmse : 0.0967



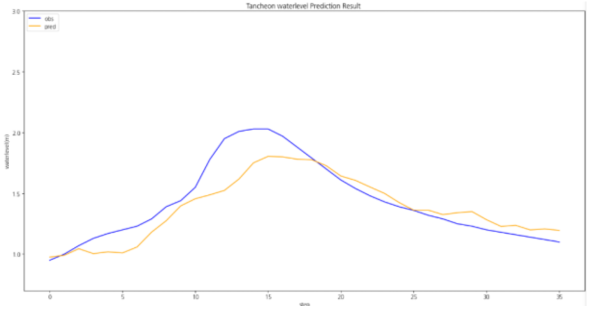
testdata_ 240723



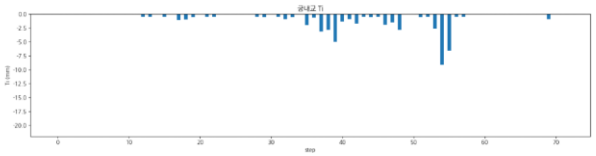
testdata_ 240723



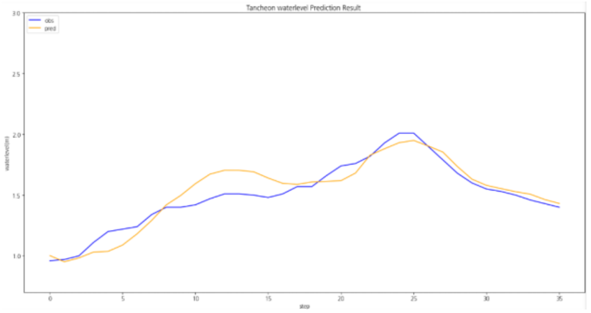
testdata_ 250716



testdata_ 250716



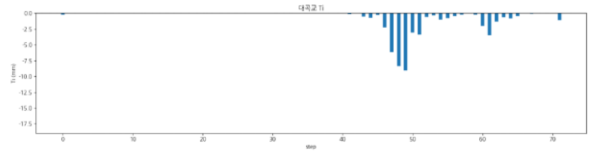
testdata_ 250814



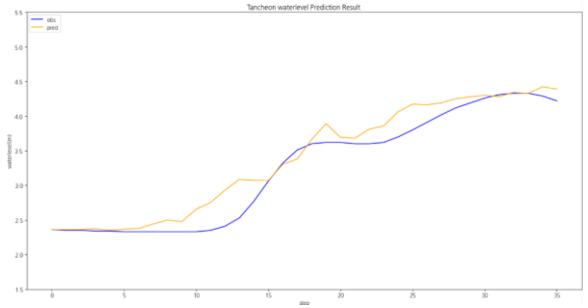
testdata_250814

누적 강우 110mm 이상 조합 찾기 5
예측 코드 – 대곡교 1

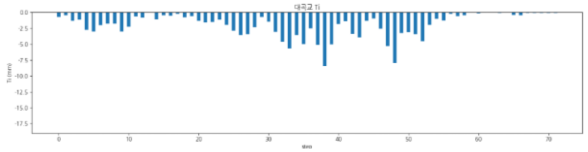
	대곡교
testdata_200802	r2 : 0.9246 mse : 0.0460 mae : 0.1541 rmse : 0.2144
testdata_220712	r2 : -0.1000 mse : 0.1540 mae : 0.3041 rmse : 0.3924
testdata_230711	r2 : 0.8276 mse : 0.1568 mae : 0.3195 rmse : 0.3961



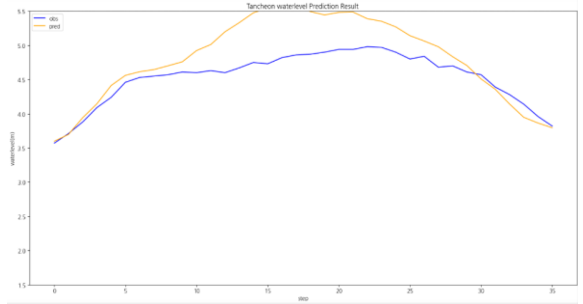
testdata_200802



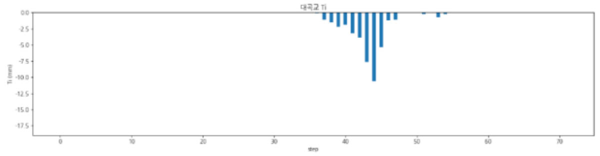
testdata_ 200802



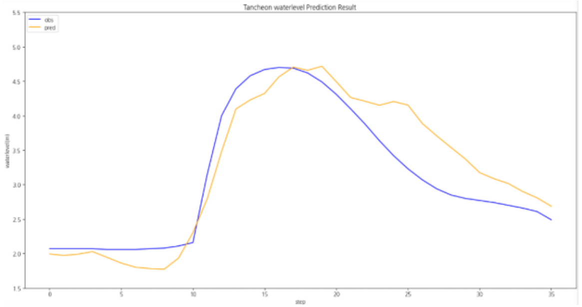
testdata_ 220712



testdata_ 220712



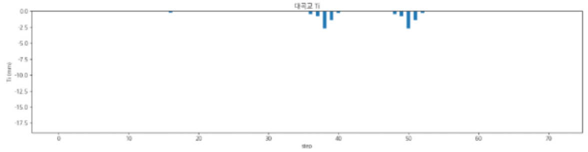
testdata_ 230711



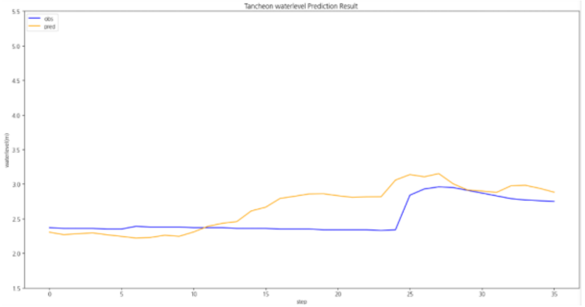
testdata_ 230711

누적 강우 110mm 이상 조합 찾기 5
예측 코드 – 대곡교 2

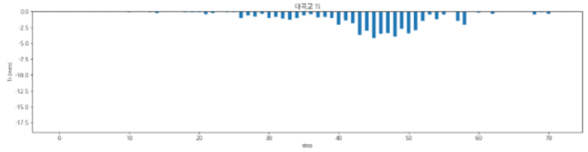
	대곡교
testdata_240723	r2 : -0.551 mse : 0.1258 mae : 0.2542 rmse : 0.3547
testdata_250716	r2 : 0.9577 mse : 0.0195 mae : 0.1145 rmse : 0.1397
testdata_250814	r2 : 0.6115 mse : 0.1348 mae : 0.3007 rmse : 0.3671



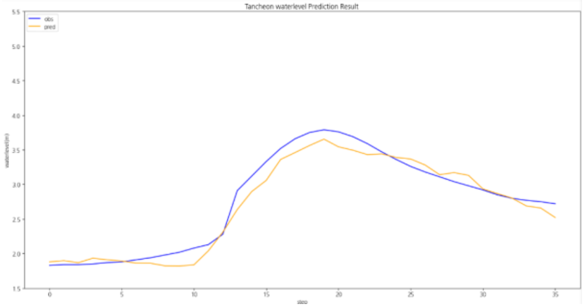
testdata_240723



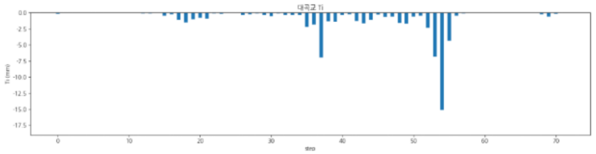
testdata_240723



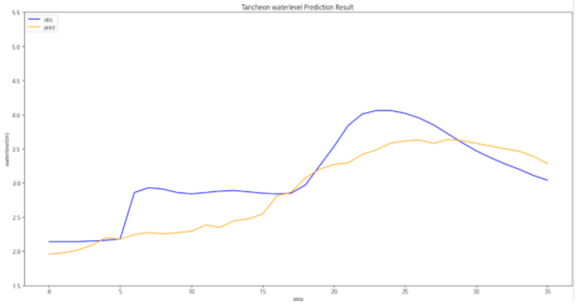
testdata_250716



testdata_250716



testdata_250814



testdata_250814

1-6. 누적 강우 110mm 이상 조합 찾기

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우 110mm 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

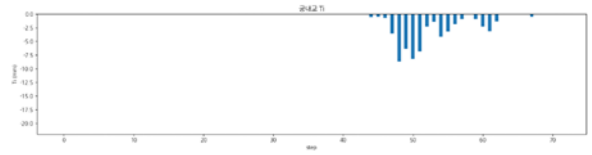
학습 : 총 11개 中 9개 데이터 파일 학습에 사용 (검증 2개)

검증 : 학습 데이터에서 제외한 2개 + 4개 이벤트 사용

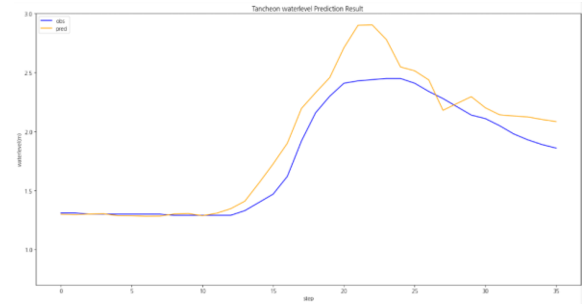
- 학습 : 2018 ~ 2024 (8개)
- 검증 : 2014(1개)
- 테스트 : 2017 (1개)

누적 강우 110mm 이상 조합 찾기 6
예측 코드 – 궁내교 1

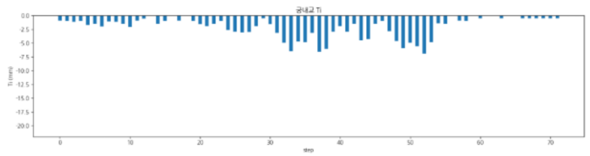
	궁내교
testdata_200802	r2 : 0.8438 mse : 0.0330 mae : 0.1330 rmse : 0.1817
testdata_220712	r2 : -0.9135 mse : 0.0759 mae : 0.2395 rmse : 0.2755
testdata_230711	r2 : 0.8268 mse : 0.0484 mae : 0.1442 rmse : 0.2201



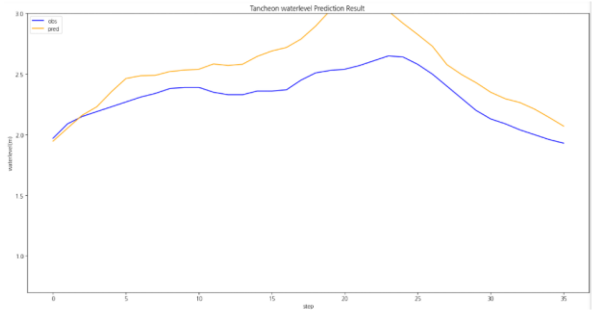
testdata_ 200802



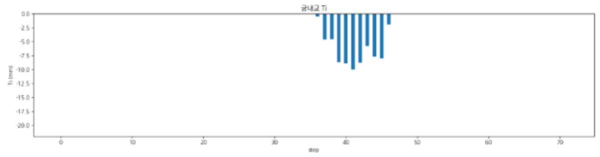
testdata_ 200802



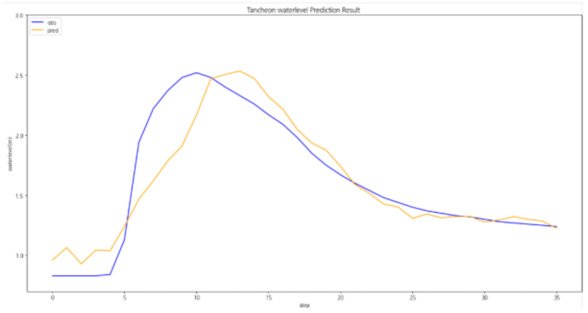
testdata_ 220712



testdata_ 220712



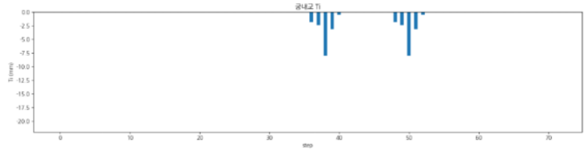
testdata_230711



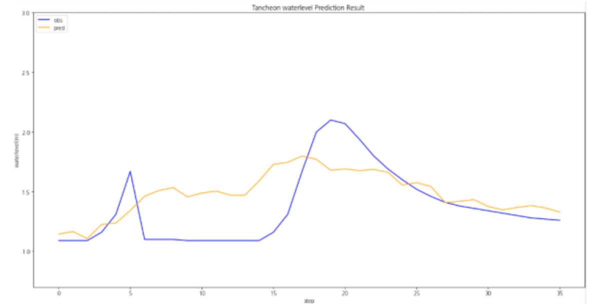
testdata_230711

누적 강우 110mm 이상 조합 찾기 6
예측 코드 – 궁내교 2

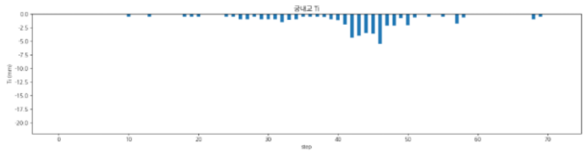
	궁내교
testdata_240723	r2 : 0.2097 mse : 0.0736 mae : 0.2087 rmse : 0.2713
testdata_250716	r2 : 0.6942 mse : 0.0306 mae : 0.1282 rmse : 0.1751
testdata_250814	r2 : 0.8607 mse : 0.0098 mae : 0.0703 rmse : 0.0990



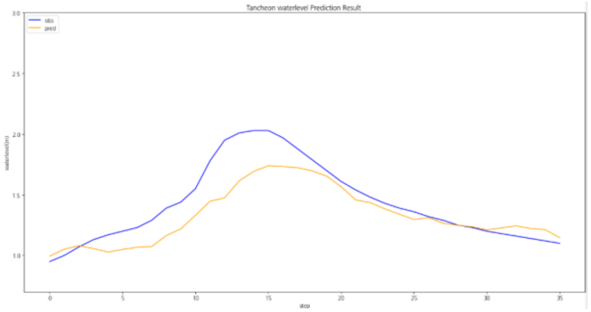
testdata_ 240723



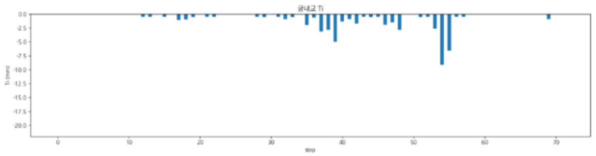
testdata_ 240723



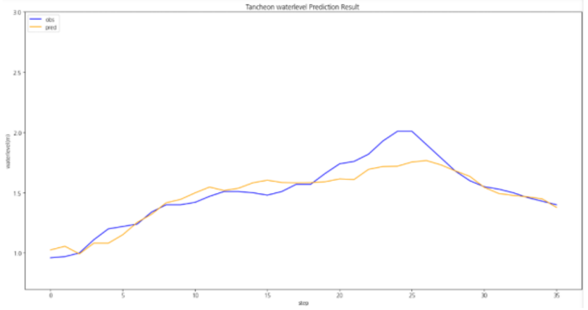
testdata_ 250716



testdata_ 250716



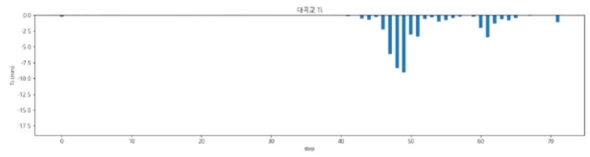
testdata_ 250814



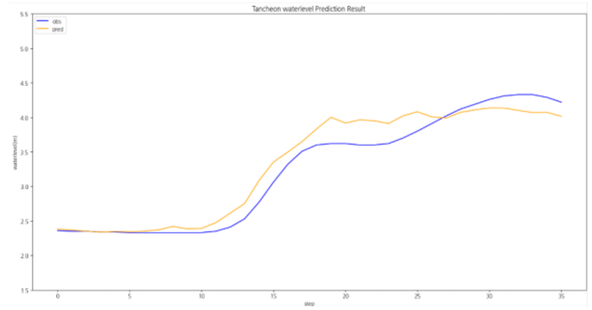
testdata_250814

누적 강우 110mm 이상 조합 찾기 6
예측 코드 – 대곡교 1

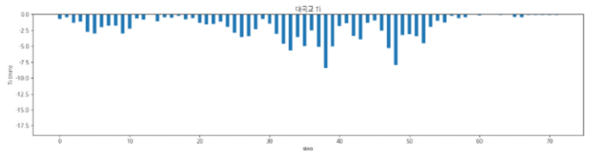
	대곡교
testdata_200802	r2 : 0.9343 mse : 0.0400 mae : 0.1605 rmse : 0.2000
testdata_220712	r2 : 0.0911 mse : 0.1272 mae : 0.3334 rmse : 0.3567
testdata_230711	r2 : 0.8255 mse : 0.1587 mae : 0.3229 rmse : 0.3984



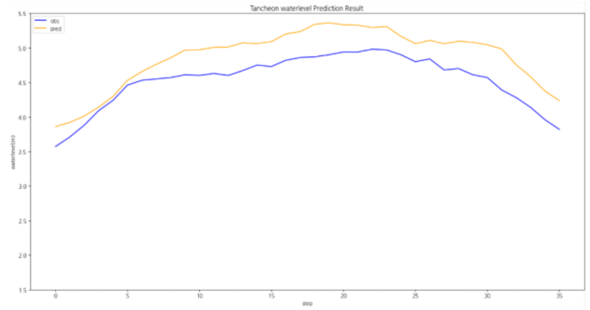
testdata_200802



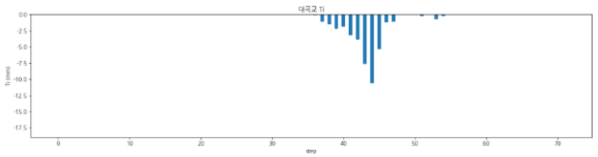
testdata_ 200802



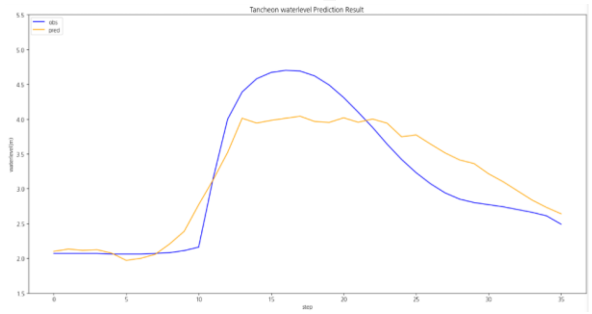
testdata_ 220712



testdata_ 220712



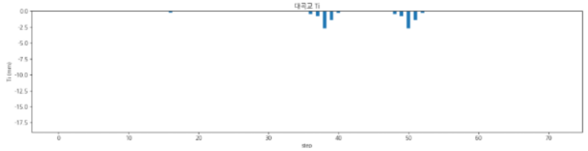
testdata_ 230711



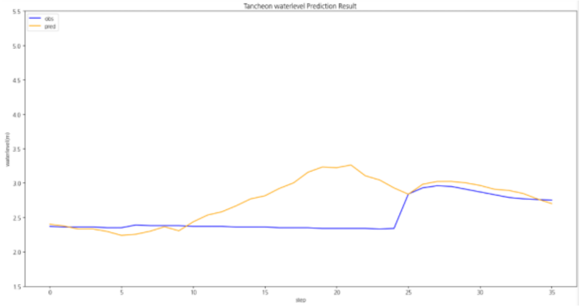
testdata_ 230711

누적 강우 110mm 이상 조합 찾기 6
예측 코드 – 대곡교 2

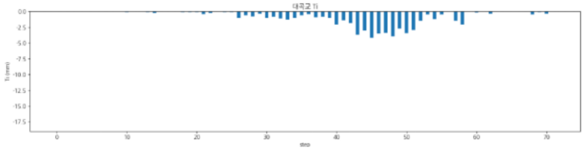
	대곡교
testdata_240723	r2 : -2.078 mse : 0.1644 mae : 0.2689 rmse : 0.4054
testdata_250716	r2 : 0.8521 mse : 0.0682 mae : 0.2196 rmse : 0.2613
testdata_250814	r2 : 0.7653 mse : 0.0814 mae : 0.2219 rmse : 0.2853



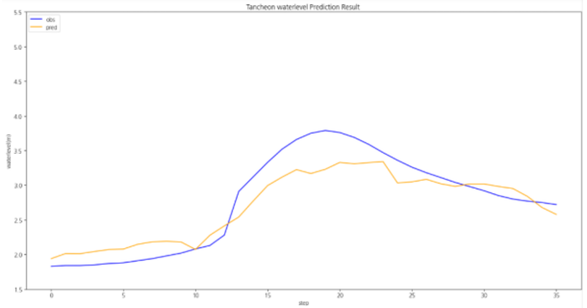
testdata_240723



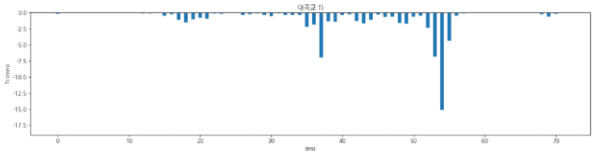
testdata_240723



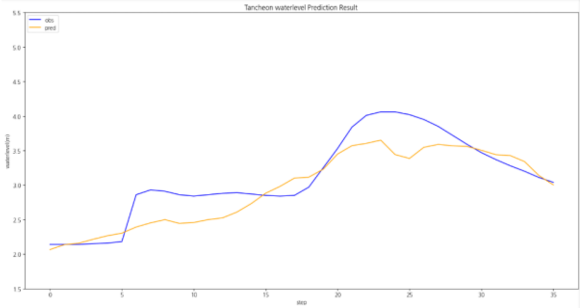
testdata_250716



testdata_250716



testdata_250814



testdata_250814

결론

현재 조합 중에는 4 (피크 재현 + 안정성 가장 좋음) ,
5 (일부 음수 발생, 전체 평균은 안정) 가 나음

* model_(gn/dg)_wl18

2. 관심 수위 이상

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 67개 中 64개 데이터 파일 학습에 사용 (검증 3개)

검증 : 학습 데이터에서 제외한 3개 + 학습에서 검증에 사용된 2개 이벤트 사용

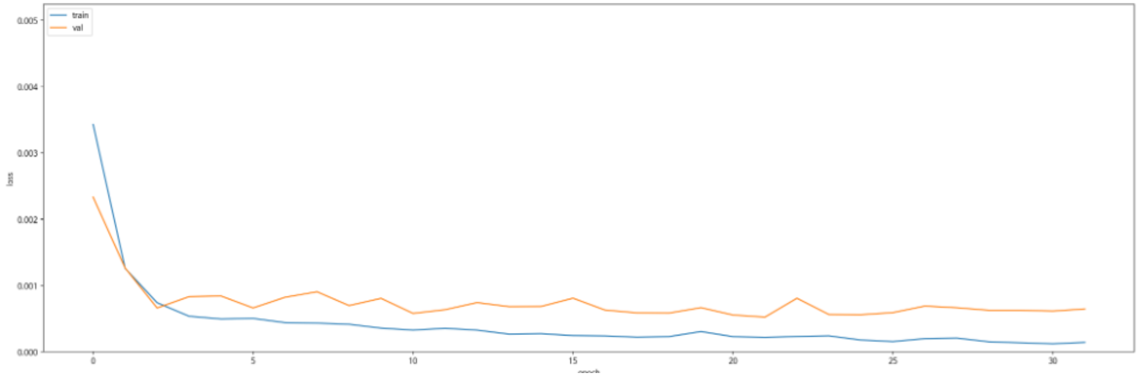
- 학습 : 2014 ~ 20230629 (45개)
- 검증 : 20230713 ~ 250813 (13개)
- 테스트 : 2550916 ~ 251006 (6개)

관심 수위 이상
예측 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

일괄 예측 수행 결과

Train Chart



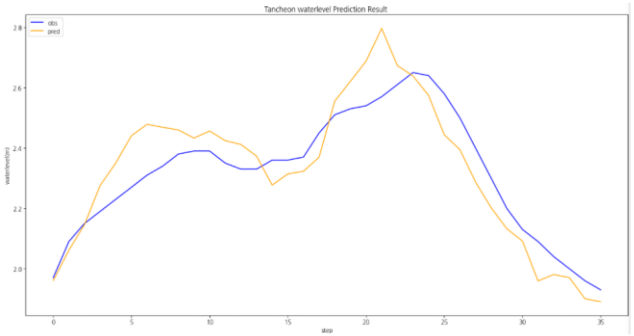
Train (5065 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0398	0.0274	0.0543	0.986375	0.986791	2.606402

Valid (1346 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0667	0.0513	0.0934	0.942399	0.946896	5.359824

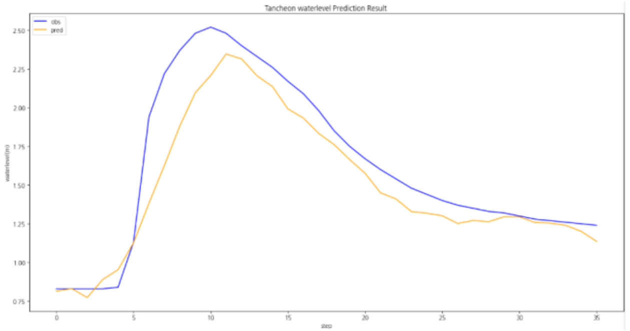
Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0449	0.0328	0.0718	0.950739	0.958120	4.395504

관심 수위 이상 예측 코드 - 궁내교

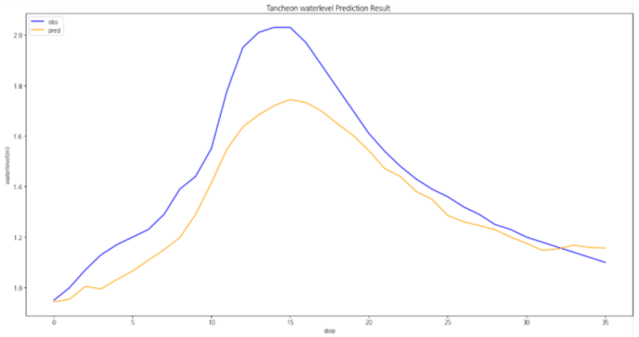
	궁내교
testdata_220712	r2 : 0.7777 mse : 0.0088 mae : 0.0801 rmse : 0.0939
testdata_230711	r2 : 0.8560 mse : 0.0402 mae : 0.1376 rmse : 0.2006
testdata_250716	r2 : 0.7887 mse : 0.0211 mae : 0.1130 rmse : 0.1455
testdata_250814	r2 : 0.6514 mse : 0.0245 mae : 0.1184 rmse : 0.1567



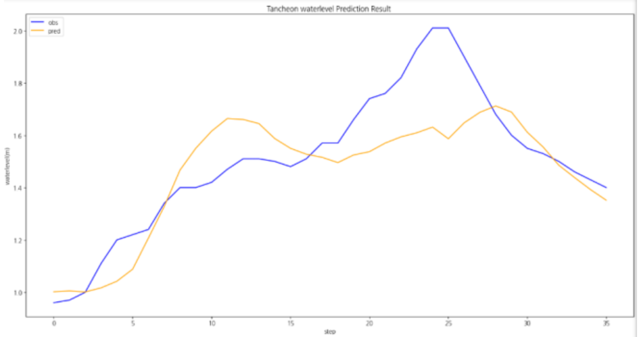
testdata_220712



testdata_230711



testdata_250716



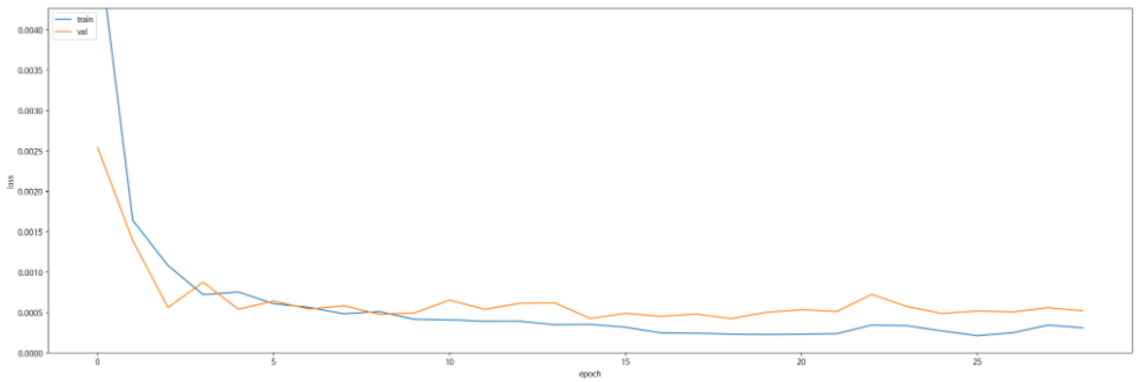
testdata_250814

* model_(gn/dg)_wl18

관심 수위 이상 예측 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

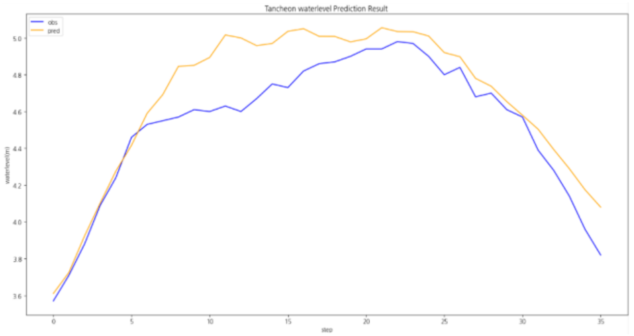
Train (5065 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0643	0.0226	0.0959	0.987721	0.989193	5.235190

Valid (1346 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0900	0.0333	0.1277	0.955670	0.956173	2.551442

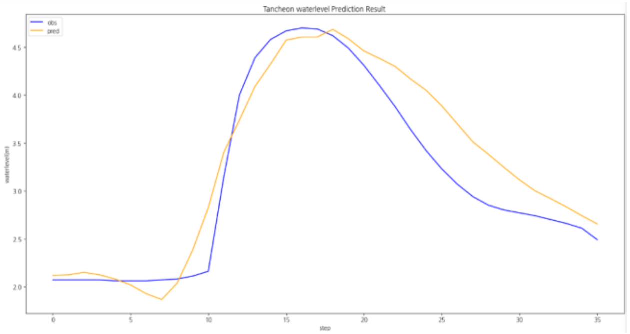
Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0700	0.0260	0.1105	0.952298	0.953230	3.660785

관심 수위 이상
예측 코드 - 대곡교

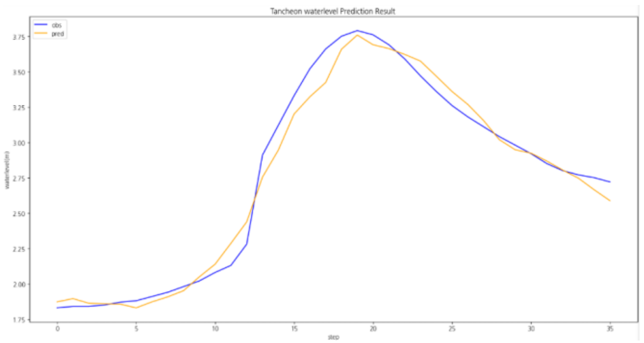
	대곡교
testdata_220712	r2 : 0.7789 mse : 0.0309 mae : 0.1394 rmse : 0.1759
testdata_230711	r2 : 0.8833 mse : 0.1061 mae : 0.2572 rmse : 0.3258
testdata_250716	r2 : 0.9808 mse : 0.0088 mae : 0.0723 rmse : 0.0939
testdata_250814	r2 : 0.8550 mse : 0.0503 mae : 0.1777 rmse : 0.2242



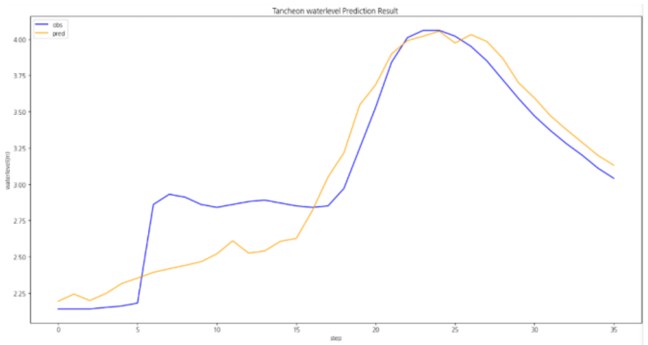
testdata_220712



testdata_230711



testdata_250716



testdata_250814

* model_(gn/dg)_wl18_2

2-2. 관심 수위 이상 Testdata 증강

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0001
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 67개 中 61개 데이터 파일 학습에 사용 (검증 6개)

검증 : 학습 데이터에서 제외한 6개 + 학습에서 검증에 사용된 2개 이벤트 사용

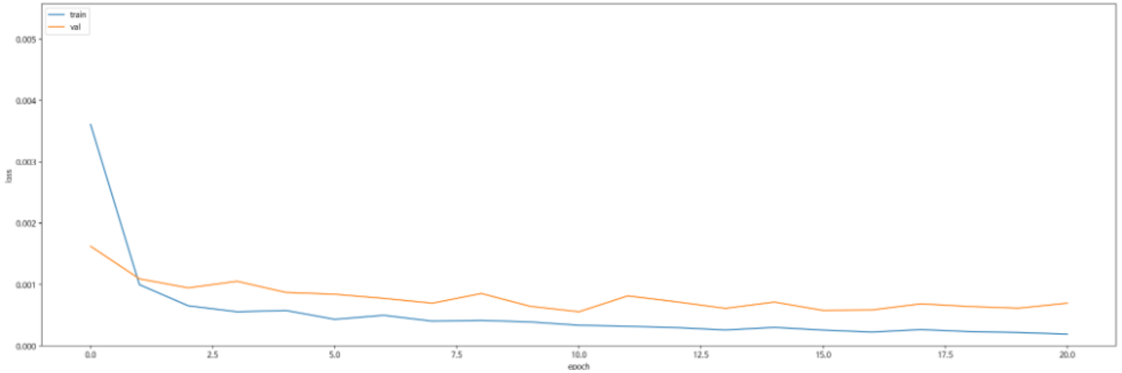
- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

* model_(gn/dg)_wl18_2

관심 수위 이상 + testdata 증강 예측 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

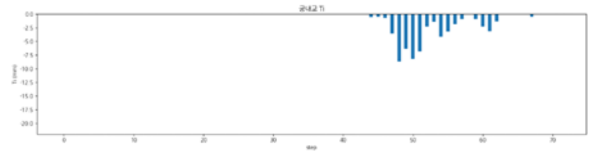
Train (4937 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0481	0.0322	0.0686	0.978650	0.979667	6.814979

Valid (1249 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0708	0.0549	0.0963	0.941805	0.947482	5.596691

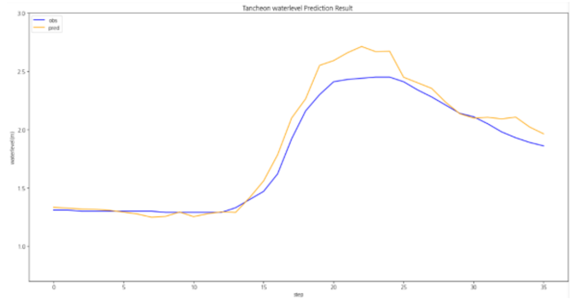
Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0472	0.0343	0.0766	0.943811	0.949195	6.698679

관심 수위 이상 + testdata 증강 예측 코드 – 궁내교 1

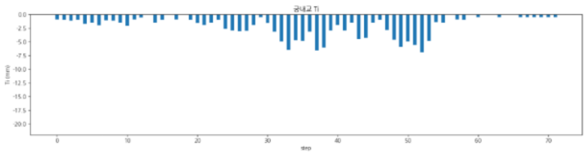
	궁내교
testdata_200802	r2 : 0.9353 mse : 0.0136 mae : 0.0837 rmse : 0.1169
testdata_220712	r2 : 0.2652 mse : 0.0291 mae : 0.1272 rmse : 0.1707
testdata_230711	r2 : 0.8690 mse : 0.0366 mae : 0.1252 rmse : 0.1914



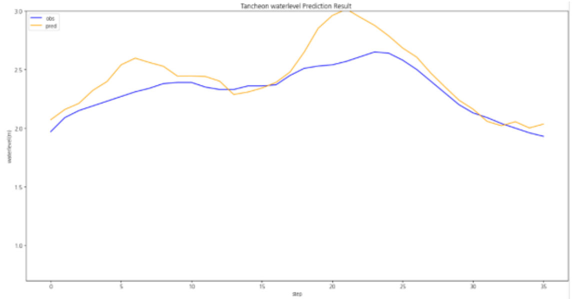
testdata_ 200802



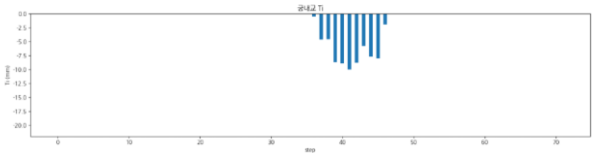
testdata_ 200802



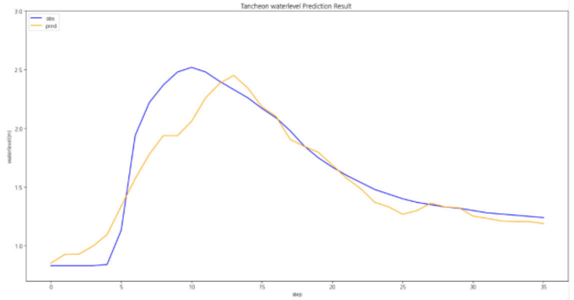
testdata_ 220712



testdata_ 220712



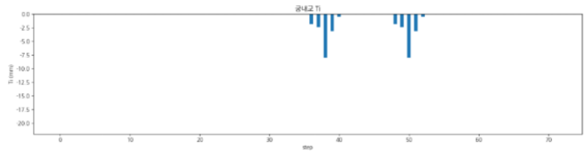
testdata_230711



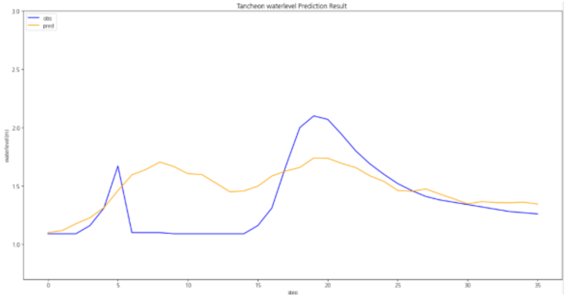
testdata_230711

관심 수위 이상 + testdata 증강 예측 코드 – 궁내교 2

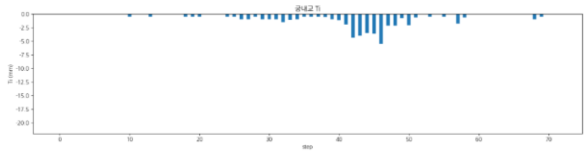
	궁내교
testdata_240723	r2 : 0.1206 mse : 0.0819 mae : 0.2115 rmse : 0.2862
testdata_250716	r2 : 0.7039 mse : 0.0296 mae : 0.1295 rmse : 0.1723
testdata_250814	r2 : 0.4391 mse : 0.0395 mae : 0.1522 rmse : 0.1988



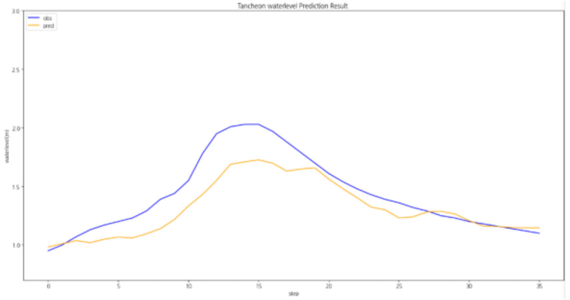
testdata_ 240723



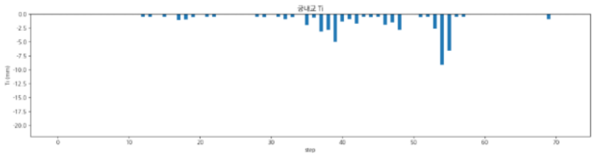
testdata_ 240723



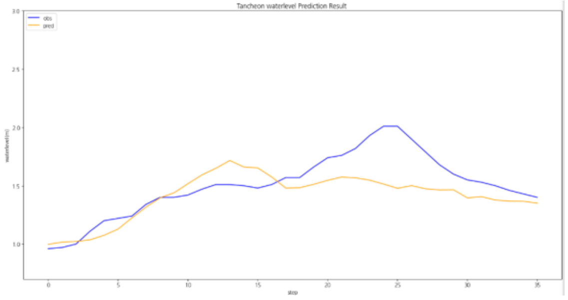
testdata_ 250716



testdata_ 250716



testdata_ 250814



testdata_250814

* model_(gn/dg)_wl18_2

관심 수위 이상 + testdata 증강 예측 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

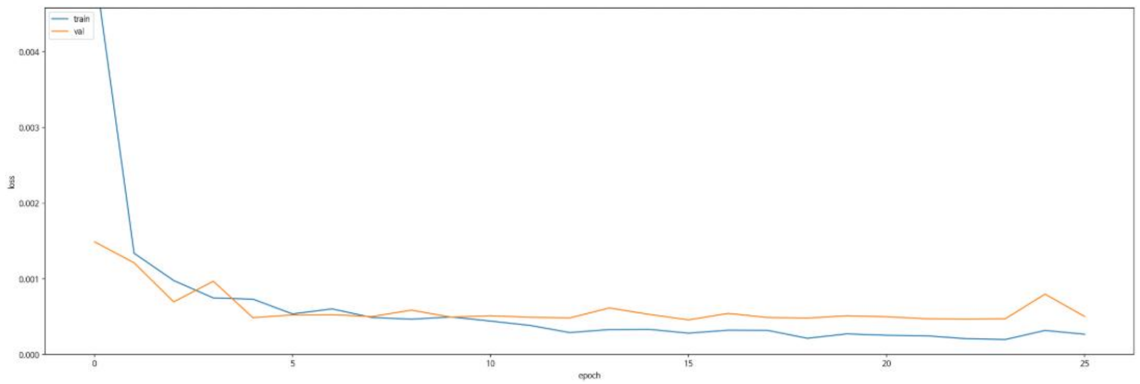
일괄 예측 수행 결과

Train (4937 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0780	0.0275	0.1126	0.983367	0.983731	6.069073

Valid (1249 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0972	0.0362	0.1324	0.955390	0.955707	3.890129

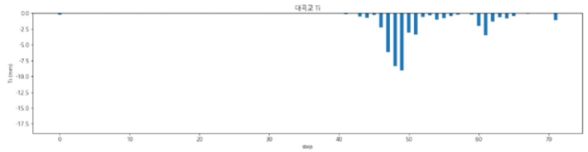
Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0829	0.0307	0.1280	0.936055	0.938276	6.933598

Train Chart

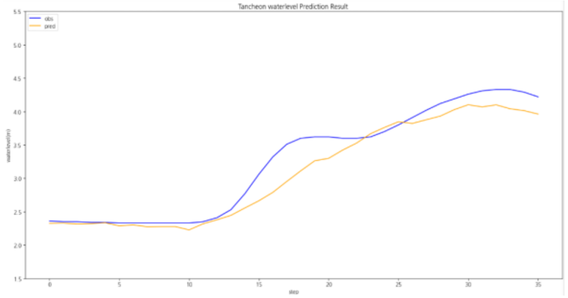


관심 수위 이상 + testdata 증강
예측 코드 – 대곡교 1

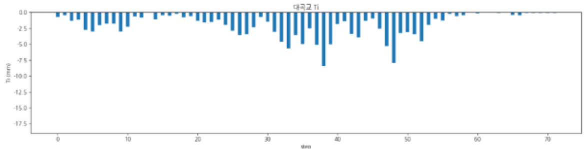
	대곡교
testdata_200802	r2 : 0.9182 mse : 0.0498 mae : 0.1637 rmse : 0.2233
testdata_220712	r2 : 0.4487 mse : 0.0771 mae : 0.2441 rmse : 0.2778
testdata_230711	r2 : 0.8872 mse : 0.1026 mae : 0.2578 rmse : 0.3203



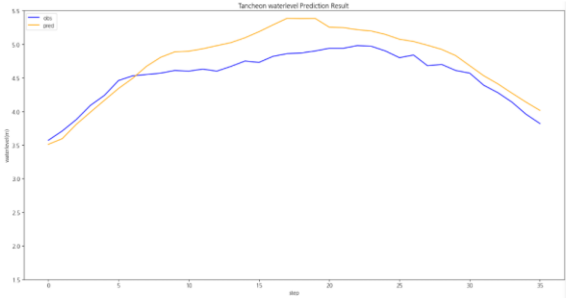
testdata_200802



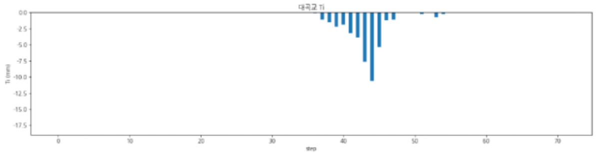
testdata_ 200802



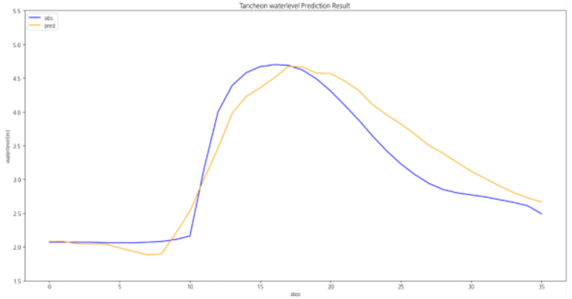
testdata_ 220712



testdata_ 220712



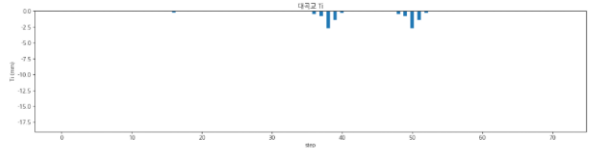
testdata_ 230711

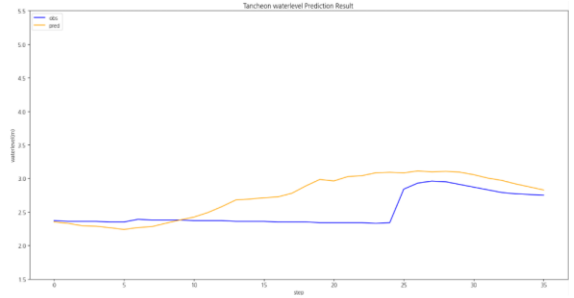


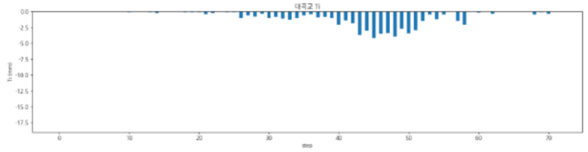
testdata_ 230711

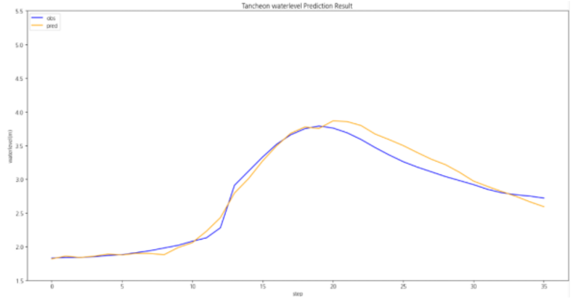
관심 수위 이상 + testdata 증강
예측 코드 – 대곡교 2

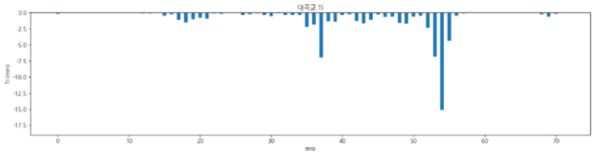
	대곡교
testdata_240723	r2 : -1.2187 mse : 0.1185 mae : 0.2587 rmse : 0.3442
testdata_250716	r2 : 0.9720 mse : 0.0129 mae : 0.0860 rmse : 0.1136
testdata_250814	r2 : 0.8273 mse : 0.0599 mae : 0.1907 rmse : 0.2448

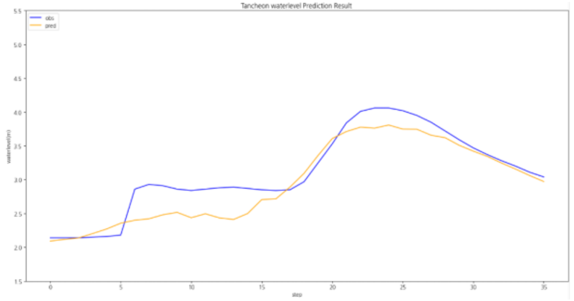

testdata_240723


testdata_240723


testdata_250716


testdata_250716


testdata_250814


testdata_250814

* model_(gn/dg)_wl18_3

2-3. 관심 수위 이상 Testdata 증강 MAE + 0.0005

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0005
- Loss_function = MAE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

학습데이터

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 67개 中 61개 데이터 파일 학습에 사용 (검증 6개)

검증 : 학습 데이터에서 제외한 6개 + 학습에서 검증에 사용된 2개 이벤트 사용

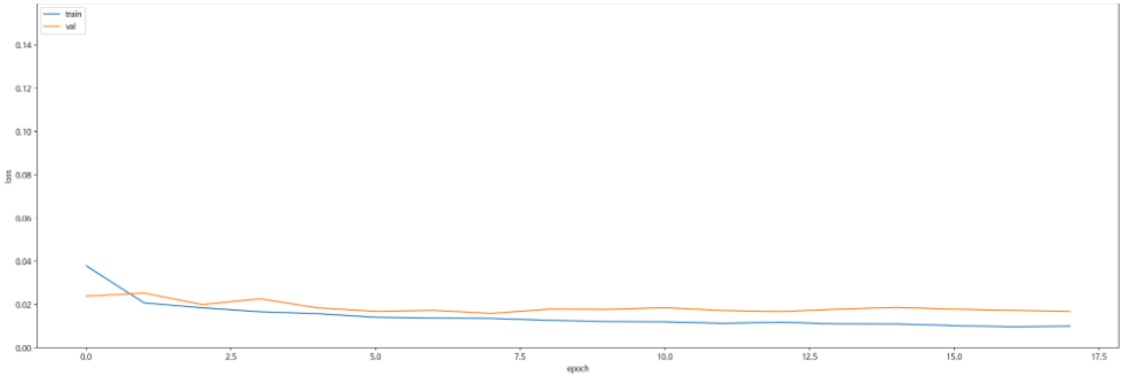
- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

* model_(gn/dg)_wl18_3

관심 수위 이상 + testdata 증강 + MAE + 0.0005 예측 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

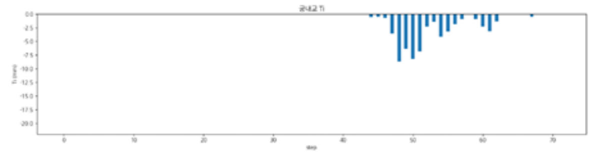
Train (5079 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0527	0.0349	0.0808	0.970485	0.971132	6.131354

Valid (1404 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0649	0.0463	0.0974	0.938086	0.942190	3.555444

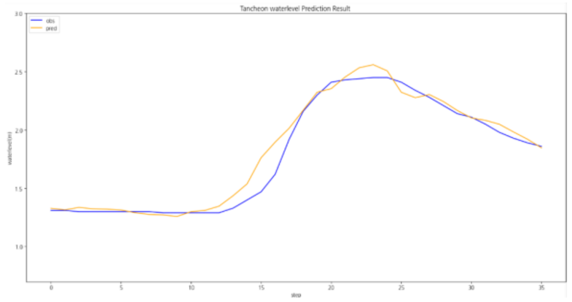
Test (297 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0403	0.0290	0.0732	0.931818	0.935694	0.423962

관심 수위 이상 + testdata 증강 + MAE + 0.0005
예측 코드 – 궁내교 1

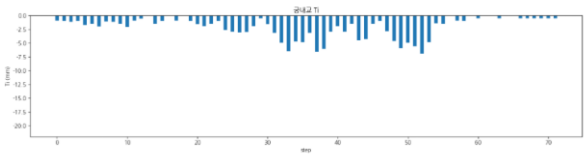
	궁내교
testdata_200802	r2 : 0.9656 mse : 0.0072 mae : 0.0558 rmse : 0.0852
testdata_220712	r2 : 0.4201 mse : 0.0230 mae : 0.1139 rmse : 0.1516
testdata_230711	r2 : 0.8622 mse : 0.0385 mae : 0.1486 rmse : 0.1963



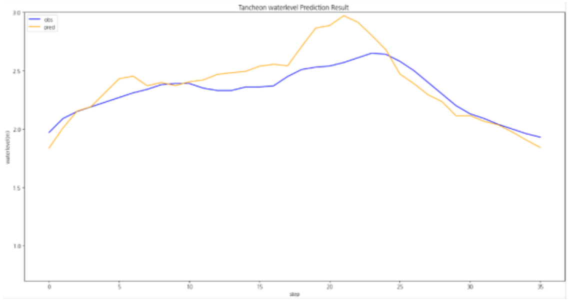
testdata_ 200802



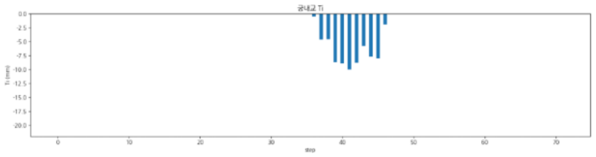
testdata_ 200802



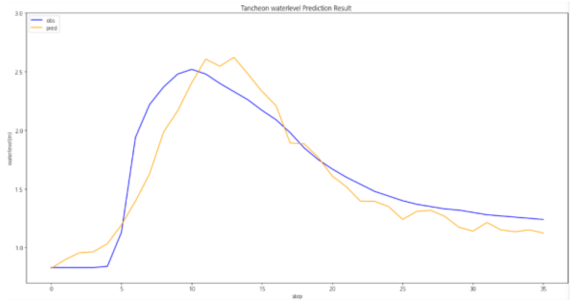
testdata_ 220712



testdata_ 220712



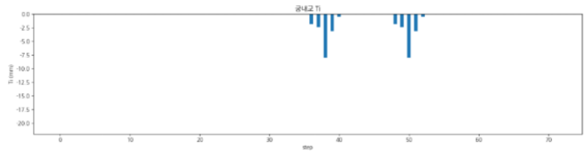
testdata_230711



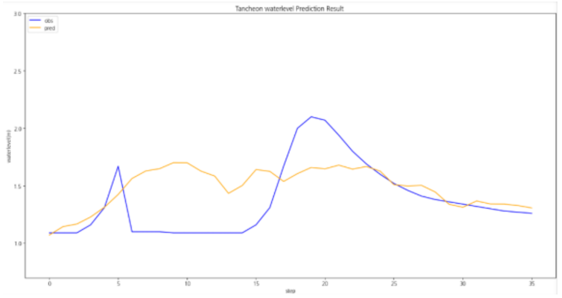
testdata_230711

관심 수위 이상 + testdata 증강 + MAE + 0.0005
예측 코드 – 궁내교 2

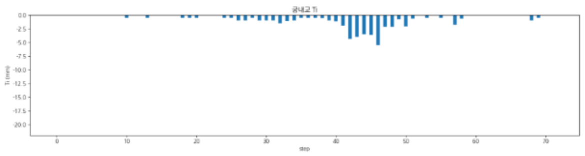
	궁내교
testdata_240723	r2 : -0.0236 mse : 0.0953 mae : 0.2271 rmse : 0.3088
testdata_250716	r2 : 0.7947 mse : 0.0205 mae : 0.1047 rmse : 0.1435
testdata_250814	r2 : 0.5087 mse : 0.0346 mae : 0.1446 rmse : 0.1861



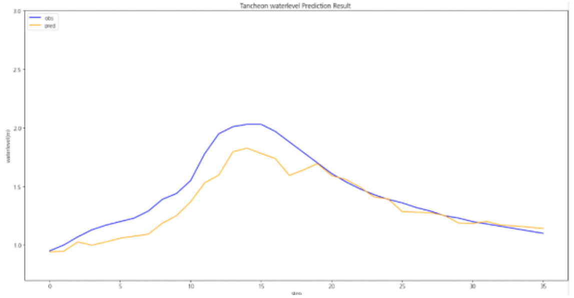
testdata_ 240723



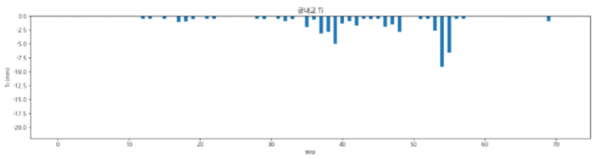
testdata_ 240723



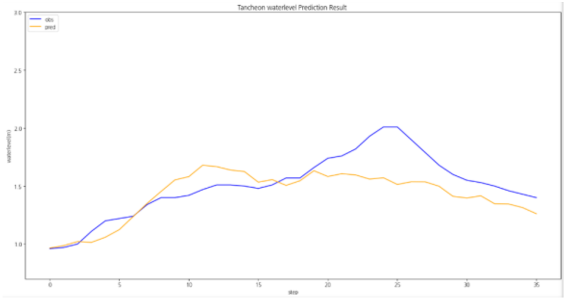
testdata_ 250716



testdata_ 250716



testdata_ 250814



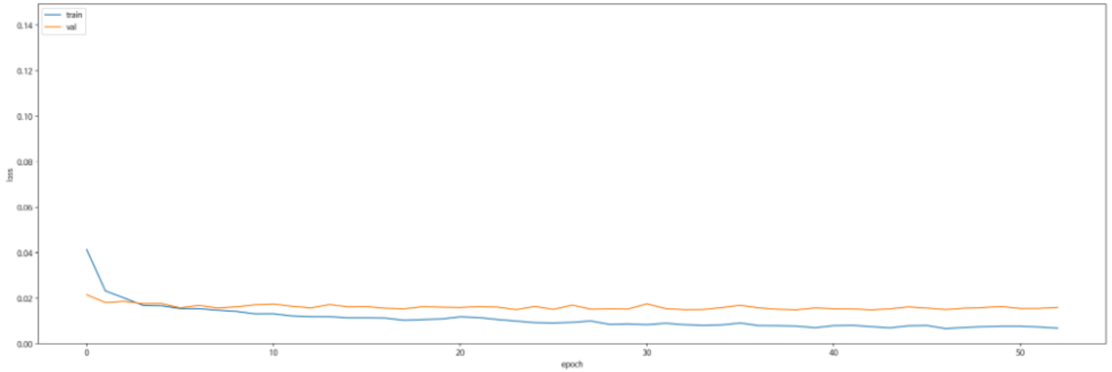
testdata_250814

* model_(gn/dg)_wl18_3

관심 수위 이상 + testdata 증강 + MAE + 0.0005 예측 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

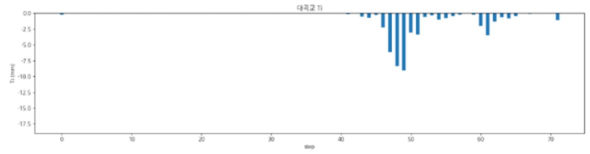
Train (5079 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0478	0.0164	0.0752	0.992509	0.992542	1.779686

Valid (1404 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0923	0.0332	0.1380	0.950117	0.951611	1.926420

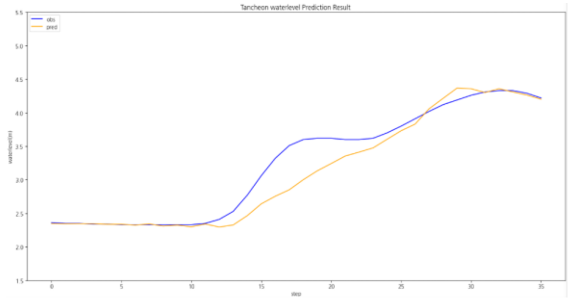
Test (297 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0502	0.0200	0.0801	0.961031	0.962256	1.172597

관심 수위 이상 + testdata 증강 + MAE + 0.0005 예측 코드 – 대곡교 1

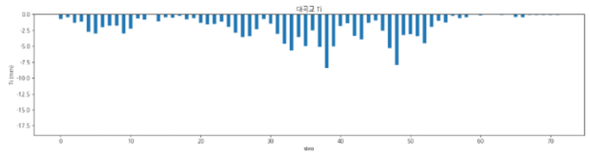
	대곡교
testdata_200802	r2 : 0.9086 mse : 0.0557 mae : 0.1438 rmse : 0.2361
testdata_220712	r2 : 0.8939 mse : 0.0148 mae : 0.0947 rmse : 0.1218
testdata_230711	r2 : 0.9252 mse : 0.0680 mae : 0.2035 rmse : 0.2608



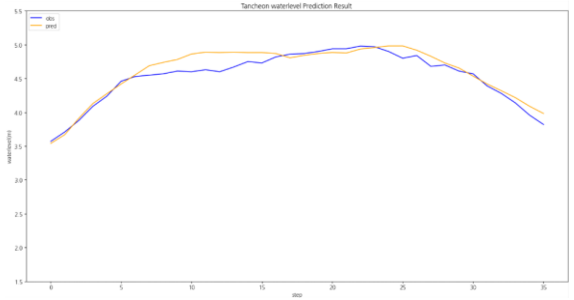
testdata_200802



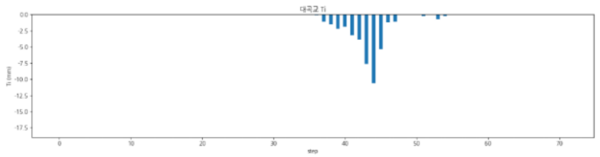
testdata_ 200802



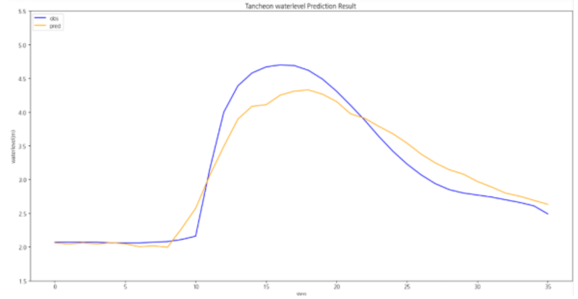
testdata_ 220712



testdata_ 220712



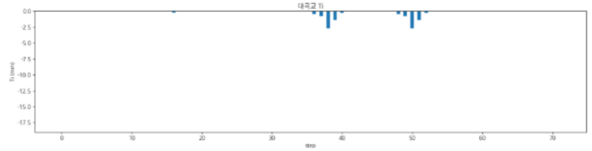
testdata_ 230711



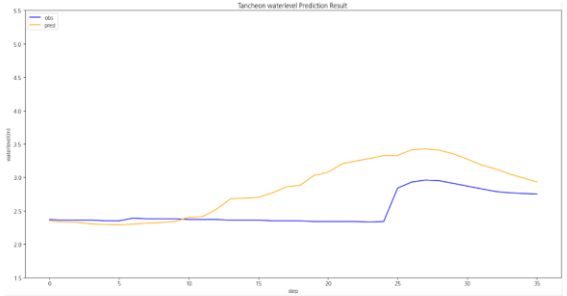
testdata_ 230711

관심 수위 이상 + testdata 증강 + MAE + 0.0005
예측 코드 – 대곡교 2

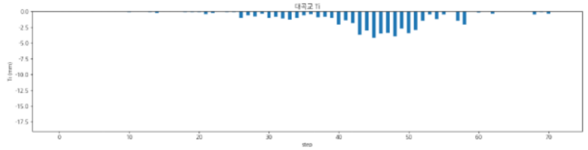
	대곡교
testdata_240723	r2 : -2.7619 mse : 0.2009 mae : 0.3459 rmse : 0.4482
testdata_250716	r2 : 0.9579 mse : 0.0194 mae : 0.0938 rmse : 0.1393
testdata_250814	r2 : 0.7833 mse : 0.0752 mae : 0.2001 rmse : 0.2742



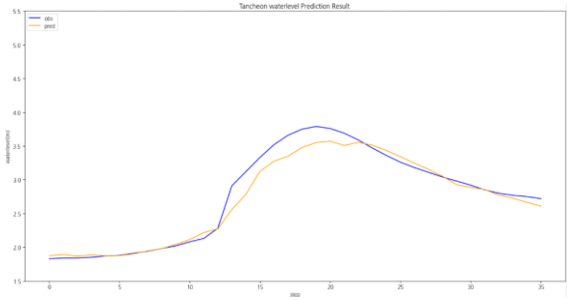
testdata_240723



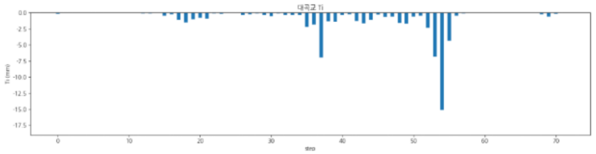
testdata_240723



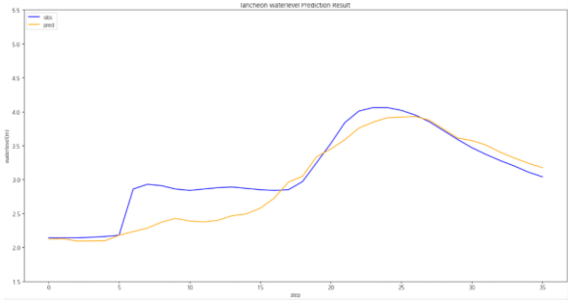
testdata_250716



testdata_250716



testdata_250814



testdata_250814

* model_(gn/dg)_msle

2-4. 관심 수위 이상 Testdata 증강 MSLE

상대오차(비율)를 더 중요하게 봄

값이 클수록 절대오차보다 비율 오차에 민감

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0001
- Loss_function = MSLE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

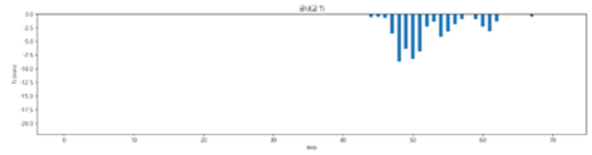
학습 : 총 67개 中 61개 데이터 파일 학습에 사용 (검증 6개)

검증 : 학습 데이터에서 제외한 6개 + 학습에서 검증에 사용된 2개 이벤트 사용

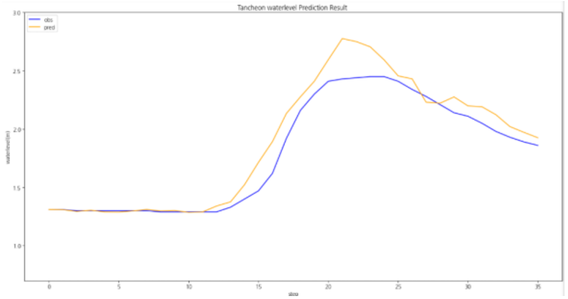
- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

관심 수위 이상 + testdata 증강 + MSLE 예측 코드 – 궁내교 1

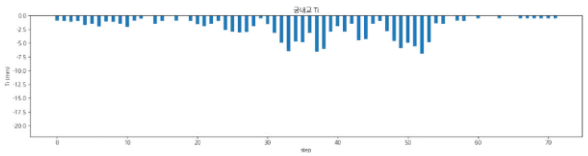
	궁내교
testdata_200802	r2 : 0.9132 mse : 0.0183 mae : 0.0952 rmse : 0.1354
testdata_220712	r2 : 0.7455 mse : 0.0100 mae : 0.0840 rmse : 0.1004
testdata_230711	r2 : 0.8928 mse : 0.0299 mae : 0.1190 rmse : 0.1731



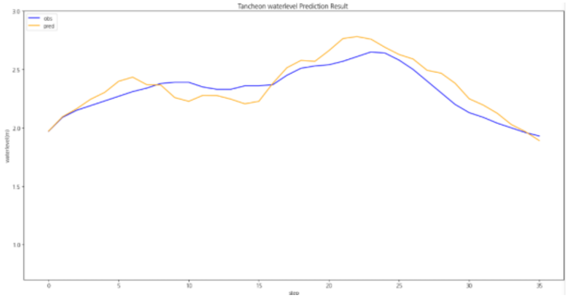
testdata_ 200802



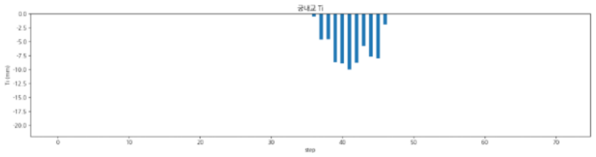
testdata_ 200802



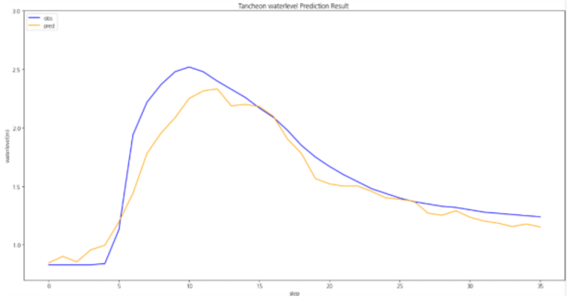
testdata_ 220712



testdata_ 220712



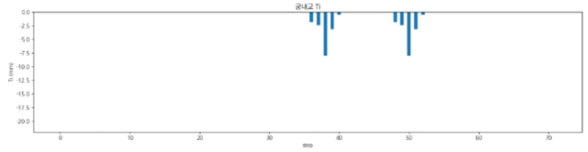
testdata_230711



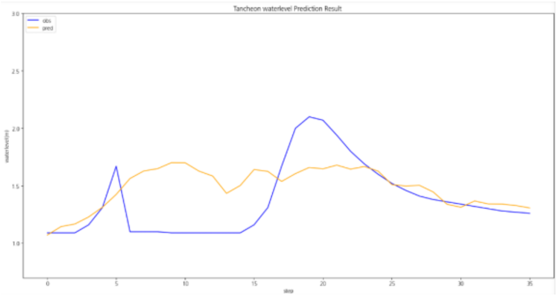
testdata_230711

관심 수위 이상 + testdata 증강 + MSLE 예측 코드 – 궁내교 2

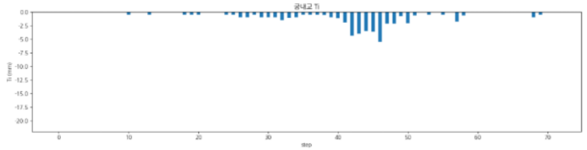
	궁내교
testdata_240723	r2 : 0.0277 mse : 0.0906 mae : 0.2251 rmse : 0.3010
testdata_250716	r2 : 0.6664 mse : 0.0334 mae : 0.1392 rmse : 0.1829
testdata_250814	r2 : 0.6655 mse : 0.0235 mae : 0.1183 rmse : 0.1535



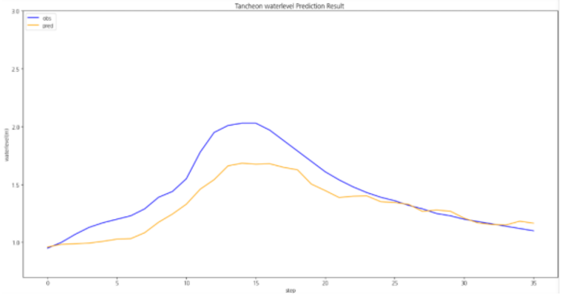
testdata_ 240723



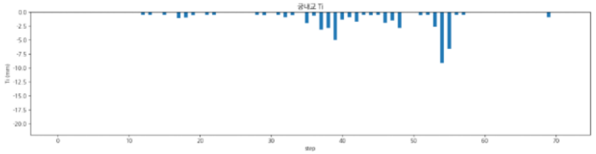
testdata_ 240723



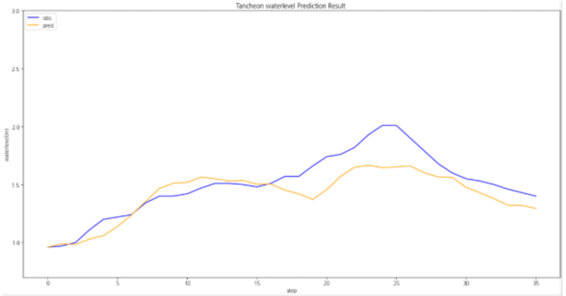
testdata_ 250716



testdata_ 250716



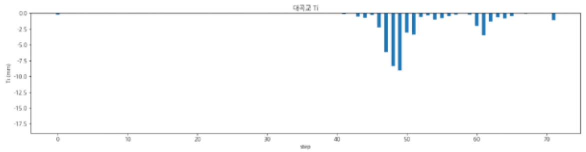
testdata_ 250814



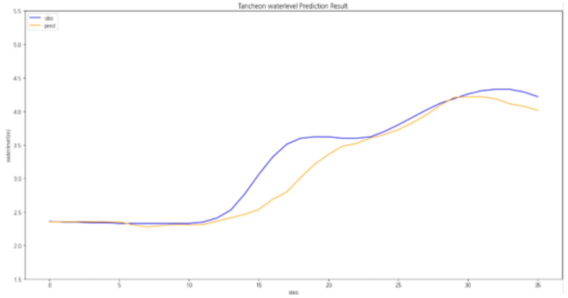
testdata_250814

관심 수위 이상 + testdata 증강 + MSLE 예측 코드 – 대곡교 1

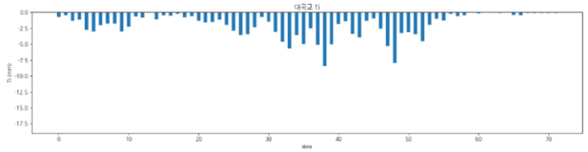
	대곡교
testdata_200802	r2 : 0.9057 mse : 0.0575 mae : 0.1462 rmse : 0.2398
testdata_220712	r2 : 0.7552 mse : 0.0342 mae : 0.1560 rmse : 0.1851
testdata_230711	r2 : 0.7906 mse : 0.1905 mae : 0.3216 rmse : 0.4364



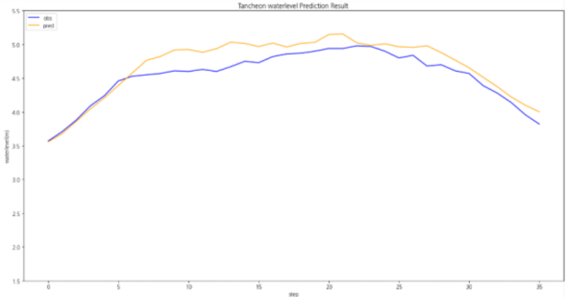
testdata_200802



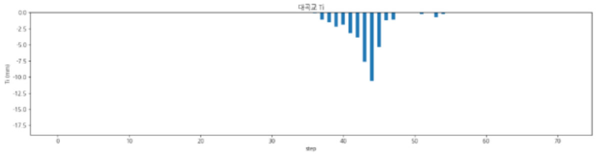
testdata_ 200802



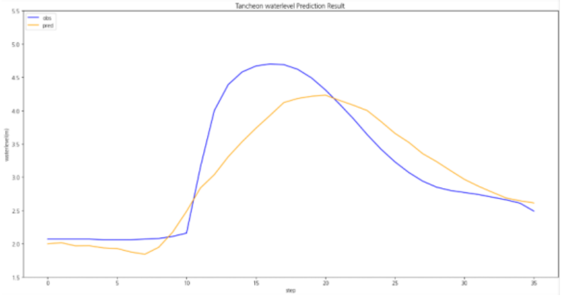
testdata_ 220712



testdata_ 220712



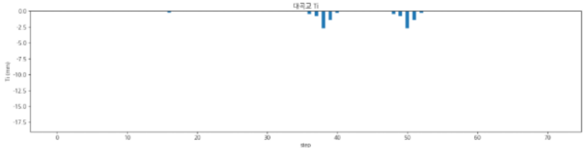
testdata_ 230711



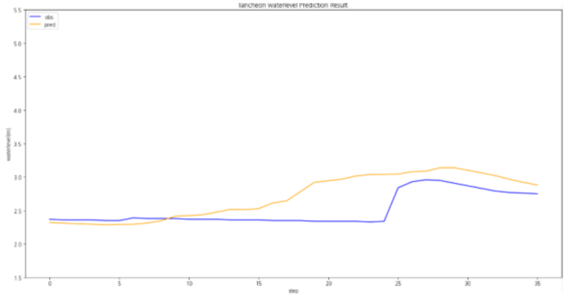
testdata_ 230711

관심 수위 이상 + testdata 증강 + MSLE 예측 코드 – 대곡교 2

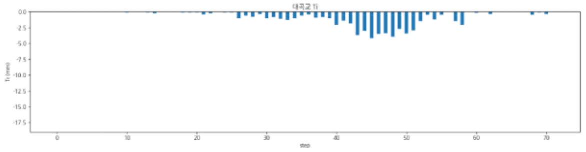
	대곡교
testdata_240723	r2 : -0.7897 mse : 0.0955 mae : 0.2293 rmse : 0.3091
testdata_250716	r2 : 0.9174 mse : 0.0381 mae : 0.1320 rmse : 0.1951
testdata_250814	r2 : 0.7456 mse : 0.0882 mae : 0.2280 rmse : 0.2971



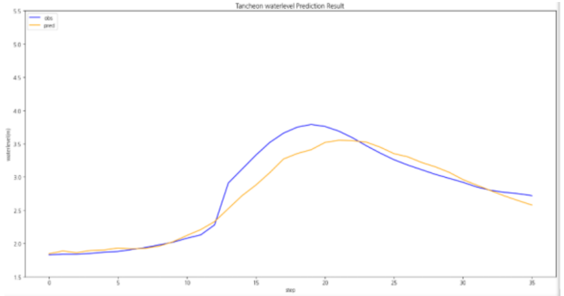
testdata_240723



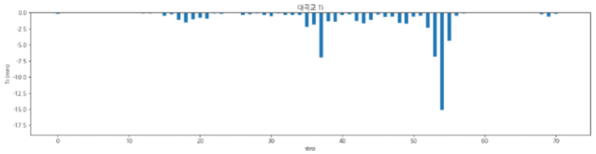
testdata_240723



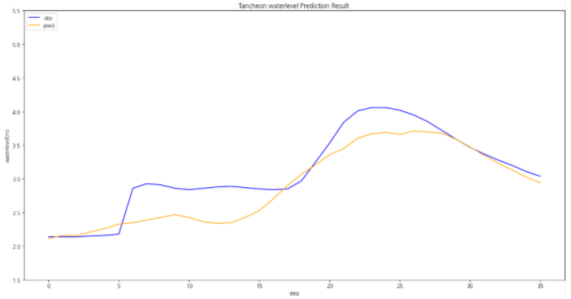
testdata_250716



testdata_250716



testdata_250814



testdata_250814

* model_(gn/dg)_huber

2-5. 관심 수위 이상 Testdata 증강 huber

작은 오차는 MSE처럼 부드럽게,

큰 오차(이상치)는 MAE처럼 강건하게

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0001
- Loss_function = Huber
- Seq_length = 36 (6시간)
- Num_of_layer = 18

학습데이터

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

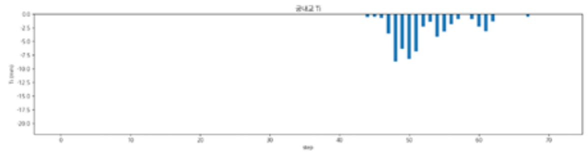
학습 : 총 67개 中 61개 데이터 파일 학습에 사용 (검증 6개)

검증 : 학습 데이터에서 제외한 6개 + 학습에서 검증에 사용된 2개 이벤트 사용

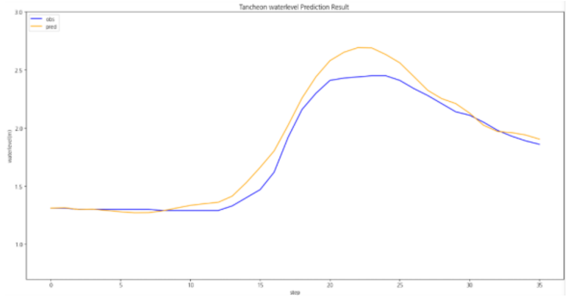
- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

관심 수위 이상 + testdata 증강 + Huber 예측 코드 – 궁내교 1

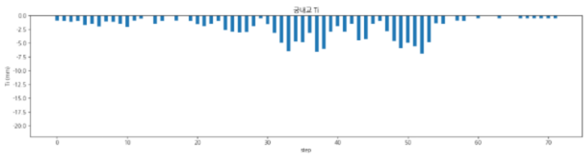
	궁내교
testdata_200802	r2 : 0.9436 mse : 0.0119 mae : 0.0800 rmse : 0.1091
testdata_220712	r2 : 0.5280 mse : 0.0187 mae : 0.1057 rmse : 0.1368
testdata_230711	r2 : 0.8739 mse : 0. 0352 mae : 0.1445 rmse : 0.1878



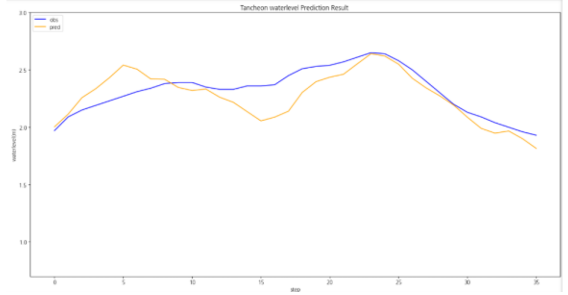
testdata_ 200802



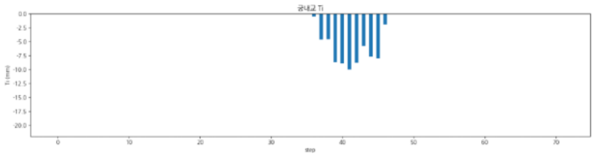
testdata_ 200802



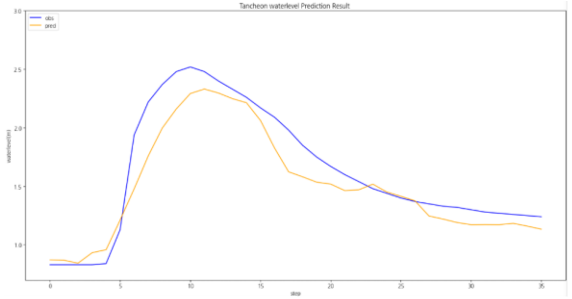
testdata_ 220712



testdata_ 220712



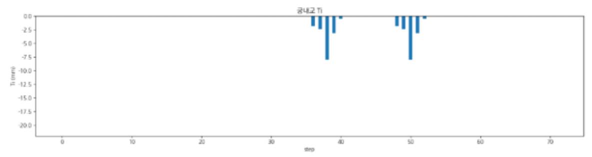
testdata_230711



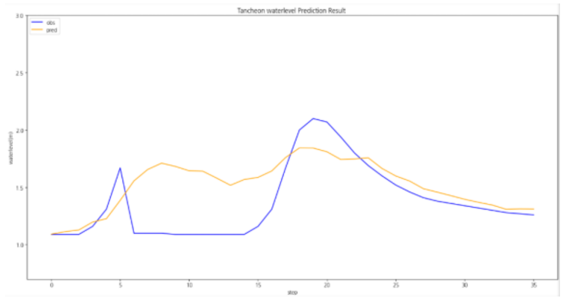
testdata_230711

관심 수위 이상 + testdata 증강 + Huber 예측 코드 – 궁내교 2

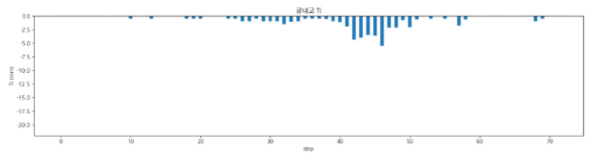
	궁내교
testdata_240723	r2 : 0.0597 mse : 0.0876 mae : 0.2157 rmse : 0.2960
testdata_250716	r2 : 0.8570 mse : 0.0143 mae : 0.0951 rmse : 0.1197
testdata_250814	r2 : 0.6947 mse : 0.0215 mae : 0.1217 rmse : 0.1467



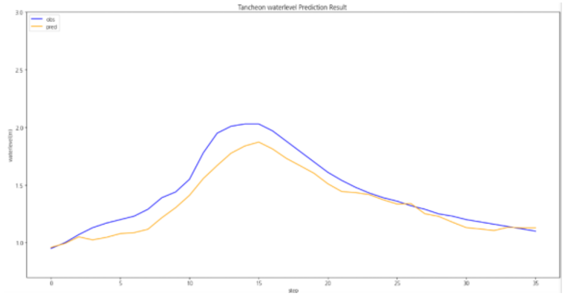
testdata_ 240723



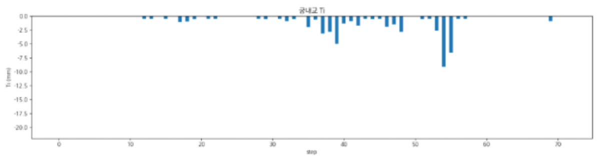
testdata_ 240723



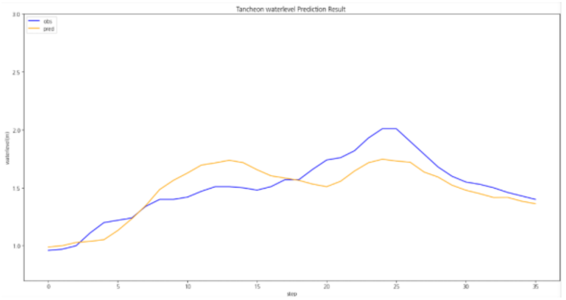
testdata_ 250716



testdata_ 250716



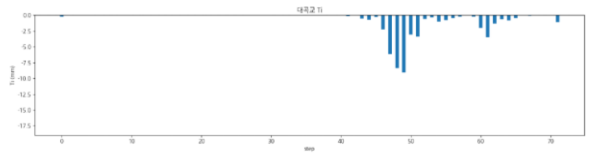
testdata_ 250814



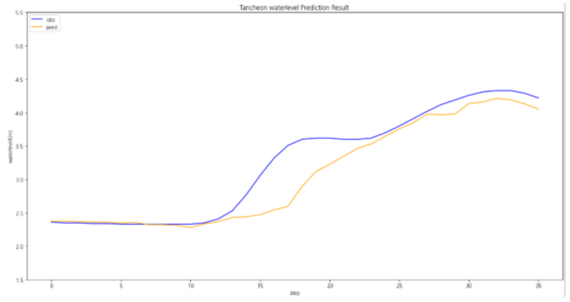
testdata_250814

관심 수위 이상 + testdata 증강 + Huber 예측 코드 – 대곡교 1

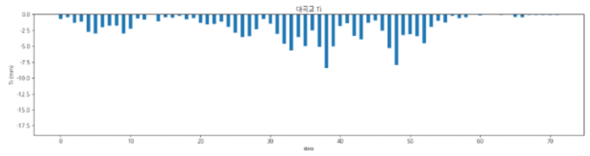
	대곡교
testdata_200802	r2 : 0.8583 mse : 0.0864 mae : 0.1800 rmse : 0.2939
testdata_220712	r2 : 0.9114 mse : 0.0123 mae : 0.0931 rmse : 0.1113
testdata_230711	r2 : 0.8354 mse : 0.1497 mae : 0.2952 rmse : 0.3870



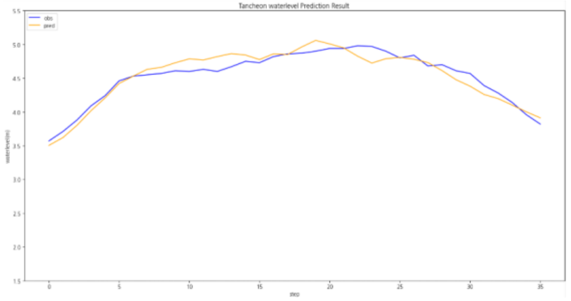
testdata_200802



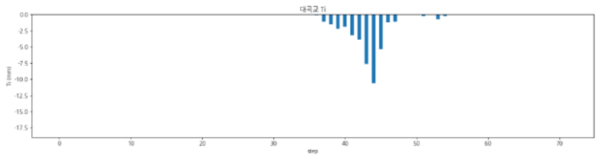
testdata_ 200802



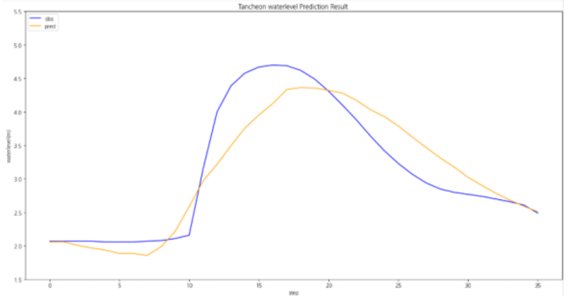
testdata_ 220712



testdata_ 220712



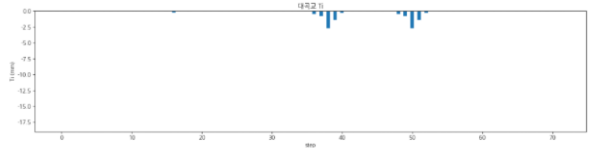
testdata_ 230711



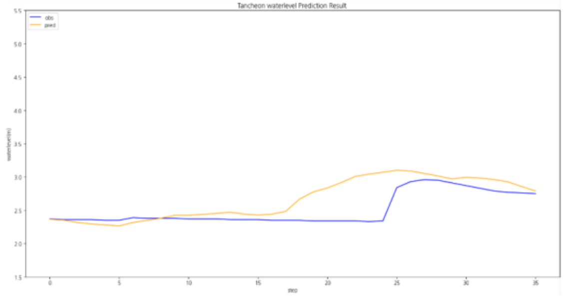
testdata_ 230711

관심 수위 이상 + testdata 증강 + Huber 예측 코드 – 대곡교 2

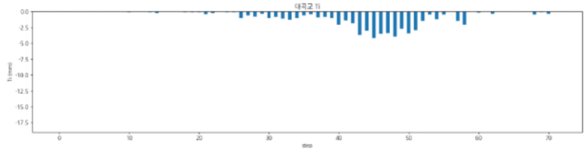
	대곡교
testdata_240723	r2 : -0.3854 mse : 0.0739 mae : 0.1788 rmse : 0.2720
testdata_250716	r2 : 0.9566 mse : 0.0200 mae : 0.1153 rmse : 0.1414
testdata_250814	r2 : 0.7191 mse : 0.0974 mae : 0.2343 rmse : 0.3122



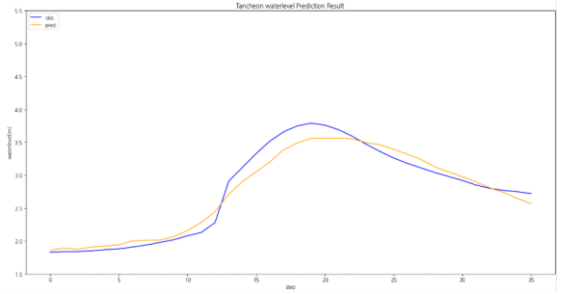
testdata_240723



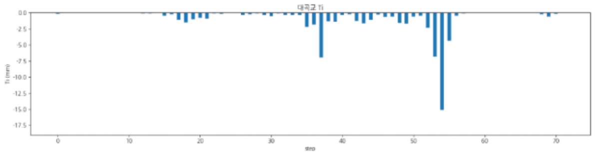
testdata_240723



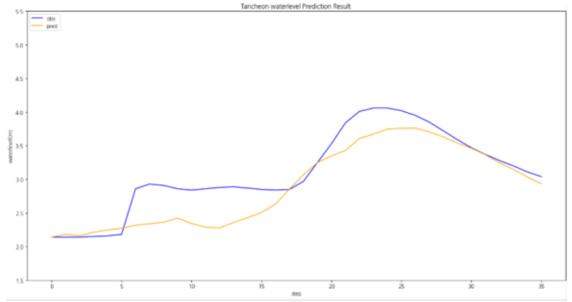
testdata_250716



testdata_250716



testdata_250814



testdata_250814

결론

관심수위이상 중에는 MAE (피크/급변 + 평균 성능 모두 최고) ,

Huber (MAE보다 이상치에 강함 , 평균 MAE는 앞 실험 보다 약간 큼) 가 나음